

A COGNITIVE THEORY OF REASONING AND CHOICE*

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We present a theory of choice in which attention to the features of options is determined by the decision maker's categorization of the current problem in a set of problems she solved in the past. Categorization depends on goal-relevant and contextual problem-level features. The model yields heterogeneity in attention and choice in a given problem based on different past experiences and instability when changes in irrelevant context cause recategorization. We show that heterogeneous and unstable representations of a choice problem unify major biases in judgment and decision making. *JEL codes*: D83, D91.

I. INTRODUCTION

Is Donald Trump a hardened criminal or the champion of ordinary Americans? Are stocks an opportunity or a gamble? Is the car rental agency Avis a loser to Hertz or an underdog worth trying? Do guns at home make you safer from potential intruders, or less safe due to the risk of an accident (Alsan, Schwartzstein, and Stantcheva 2026)?

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Economists trace choice to tastes or information. Even when these are constant, changing attention to features can alter decisions. Avis's advertising campaign "We are number two, we try harder" convinced many people to switch from Hertz to Avis not with new facts but by changing what they think about when renting a car. What shapes how we think about a choice problem? How does this, in turn, affect the choice we make?

We present a theory addressing these questions. Our broad idea is that before solving a problem, a decision maker (DM) must ask: What kind of a problem is it? What matters for solving it? This process is fundamentally cognitive. We formalize this process and its effect on choice. In the problem recognition step, the DM tunes attention to features of the choice context (location, time, etc.) to maximize the similarity between her current problem and a category of problems she solved in the past. The selected category pins down attention to the attributes of choice options (price, quality, etc.), which forms her mental representation of the problem. In the second step, the DM chooses by maximizing her objectives given the mental representation. By shaping how the DM thinks about the problem, irrelevant context and past experiences create systematic heterogeneity and instability in choice, holding tastes and information fixed.

When asked to identify the letter in the middle of a word, people recognize "A" in C\A/T and "H" in T\A/E (McClelland and Rumelhart 1981), even though the task and the target stimulus are the same. Our theory explains this instability. In the first step, the adjacent letters prompt the DM to recognize the problem as solving "which word is this?" In turn, this category of problems focuses the DM on features of the string of letters, including the visual features of "A," that allow her to fit a known word, while neglecting discrepant features that may point to a different representation. Context and experiences point to a known letter (A or H), while the ambiguous symbol "A" in isolation may bring to mind a ladder or a street sign, rather than a letter.

The same process is at play in consumer choice. A jam offered in a standard jar in a supermarket aisle prompts recognition of "buying" a staple good, focusing attention on price. The same jam offered in a fancy jar on a wooden tasting board instead prompts recognition of "consuming" pleasure, focusing attention on quality. The jam problem could be recognized in either way, depending on the location and other features, shaping attention to p or q . Three results follow.

First, changes in context cause sharp changes in valuation. A consumer recognizing choice as “buying” is insensitive to the jam quality relative to its price. By cuing a cozy breakfast experience at home, the fancy jar on the wooden board increases contextual similarity to “consuming,” inviting a switch to that category. Attention to quality then increases relative to price, raising the jam’s valuation. Price sensitivity is context-dependent because recategorization causes a discontinuous fall in attention to p relative to q .

Second, a frequently used category is more likely to be used today, because the DM endogenously focuses on the context congruent with it. A formerly poor consumer is likely to notice that the jam costs money because this feature fits her frequent poverty experiences. Reliance on “buying” then causes her to be more price elastic than a consumer without the same experiences, who relies more on “consuming” and focuses on pleasure. A “price-focused” mental set can cause the neglect of large benefits, as when modest medical co-pays reduce demand for valuable care (Baicker, Mullainathan, and Schwartzstein 2015; Chandra, Flack, and Obermeyer 2024). Experiences create heterogeneous price elasticities at constant income and tastes.

Third, a feature that is bottom-up salient, due to its prominence (Bushong et al. 2010) or contrast (Bordalo, Gennaioli, and Shleifer 2022), favors a switch to a category in which it is focal, reducing attention to other features. A consumer may neglect the risk that the jam is spoiled in normal times. Yet a public concern that a few jars are contaminated makes quality salient, favoring a switch to “consuming.” Attention is drawn to quality, including sickness from contamination, not to probabilities or price. As a result, valuation of jam falls dramatically and becomes insensitive to price. Recategorization explains how unlikely risks can be either neglected or exaggerated and leads to predictions for when each occurs.

In our theory, decision weights traditionally viewed as stable preferences (e.g., price sensitivity) or biases (e.g., probability weighting) are shaped by how the DM recognizes and represents the problem, which shapes attention to the attributes of a choice option. Section II introduces the model and shows the unifying power of this logic. Section III shows that the model generates heterogeneous and unstable price elasticities (Wakefield and Inman 2003) and mental accounting (Thaler 1980, 1999). In Section IV it accounts for judgment biases typically explained by

heuristics and their instability (Tversky and Kahneman 1974; Ba, Bohren, and Imas 2024; Bordalo et al. 2026a), as well as for narratives (Andre et al. 2021, 2022). In Section V the same logic yields, through bottom-up salience, a theory of “framing effects” (Tversky and Kahneman 1981) in riskless choice (Enke and Zimmermann 2019) and risky choice (Kahneman and Tversky 1979).

By modeling the problem recognition process, we link these biases to cognitive proxies such as contextual similarity and experiences. The hallmark of categorization is within-person multimodal attention shaped by context: a person’s attention to price relative to quality and thus her price elasticity should fall when context becomes more similar to consuming. The hallmark of experiences is reliance on familiar categories: people who experienced painful losses are likely to focus on potential losses in contextually similar problems, exhibiting risk aversion. Cognitive proxies allow a researcher to separate representations from tastes and information.

Recent work studies bottom-up salience driven by prominence (Li and Camerer 2022) and contrast (Bordalo, Gennaioli, and Shleifer 2012; Köszegi and Szeidl 2013), and top-down attention shaped by goal optimality (Sims 2003; Gabaix 2019) or priors (Schwartzstein 2014; Schwartzstein and Sunderam 2021; Gagnon, Rabin, and Schwartzstein 2023). Related approaches study insensitivity due to noise or uncertainty (Woodford 2012; Enke and Graeber 2023). Our innovation is to show how recognition affects attention top down, based on payoff-irrelevant context and experiences, and how recognition delivers framing effects through the interaction of top-down attention with bottom-up salient stimuli.

Most closely related to this article is Bordalo et al. (2026a), who document multimodality and instability in choice and attention in statistical problems. For instance, in the balls-and-urns problem (Edwards 1968), people do not integrate the base rate and the likelihood into a potentially distorted Bayesian judgment. They bimodally cluster either on the base rate or on the likelihood, with measured attention also clustering on these features. The authors propose a model of probability judgment in which attention is driven by bottom-up contrast and “prominence.” The current article generalizes this model to nonstatistical problems and micro-finds prominence through recognition.

We build on the psychology of similarity perceptions (Nosofsky 1986; Tversky 1972, 1977), top-down attention (Itti and Baldi 2009; Awh, Belopolsky, and Theeuwes 2012), and categories

(Reed 1972; Rosch and Lloyd 1978; Mack and Palmeri 2024). In Gilboa and Schmeidler (1995), Mullainathan (2002), and Fryer and Jackson (2008), the DM pays full attention and uses knowledge from categories or similar problems to resolve uncertainty over unknown attributes of current options. In our model, by contrast, there is no uncertainty and attention endogenously changes with experiences and salient stimuli, leading to choice heterogeneity and instability even in simple problems with full data. The resulting distortions then take the form of dampened or amplified sensitivity to a known feature across options, unifying base rate neglect, distorted price elasticities for known goods, and framing. Laibson (2001) models consumption habits as context-driven changes in reference points. In Samuelson (2001), a DM cocategorizes dictator and bargaining games. Neither offers a general theory of problem recognition.¹

Recent work on memory studies selective retrieval of the features of choice options (Bordalo, Gennaioli, and Shleifer 2012; Bordalo et al. 2023; Fudenberg, Lanzani, and Strack 2024, 2025; Wachter and Kahana 2024). In our model feature values are known, and memory shapes attention they receive. Future work may study the two mechanisms together.

II. THE MODEL

Section II.A describes how attention shapes valuation and choice. Section II.B describes how categorization and attention are jointly determined by recognition.

II.A. Attention and Valuation

A problem P includes (i) a nonempty finite menu O of lotteries defined on a finite probability space $(\Omega, 2^\Omega, \mathbb{P})$ and (ii) a context vector $\kappa_P = (\kappa_{P,i})_{i \in \Phi_K}$ that reports features common to all lotteries and belonging to a finite set Φ_K .

Each lottery $o \in O$ is a finite set of vectors $y = (u, e)$, called atoms. Subvector u reports the numerical values of a finite set Φ_H of payoff carriers (e.g., quality, price) that we call hedonic features, so $u \in \mathbb{R}^{\Phi_H}$. Subvector e reports the values of a finite set Φ_E of

1. In Salant and Rubinstein (2008) instability is due to the use of different choice functions for different frames. Ellis and Masatlioglu (2022) axiomatize utility that is stable within but not across categories of alternatives. Important work on categorization includes Mohlin (2014) and Jehiel (2005).

event features, where each $i \in \Phi_E$ is a partition of Ω (e.g., selected urn, color of drawn ball). Thus $e = (e_i)_{i \in \Phi_E}$ reports the cell in each partition in which u is delivered.²

To illustrate, consider a lottery delivering a jam and a monetary payoff contingent on the outcomes g or b of a wheel of fortune W . In state g , the lottery delivers a good jam, giving q_g utils and a monetary payoff m_g giving ηm_g utils, where $\eta > 0$ is the marginal utility of money. In state b it delivers a spoiled jam, giving $q_b \leq q_g$ utils and a monetary payoff m_b giving ηm_b utils. The money payoff can be the jam's price, $m_g = m_b = -p < 0$, but the example covers monetary lotteries if m_g and m_b are general and $q_g = q_b = 0$.

There are four hedonic features: the good and bad jam qualities Qg and Qb and the corresponding money payoffs Mg and Mb . Each feature takes value zero in the state where it is not delivered (e.g., Qg takes value zero in state b and q_g in g). The event feature is the wheel outcome, $e \in \{g, b\}$. Thus, atom (u, e) reports in $u \in \mathbb{R}^4$ the value of hedonic features in state e . There are two atoms: one for g , one for b .³

In expected utility (EU) theory, this description pins down valuation: the DM multiplies the total hedonics of (u, e) by the probability of e and adds across events. In our model, valuation depends also on the representation of P : a vector of attention weights $\alpha_P \in [0, 1]^\Phi$, where $\Phi = \Phi_H \cup \Phi_E \cup \Phi_K$ is the set of hedonic, event, and context features. Context weights $\alpha_{P,i}$, $i \in \Phi_K$ shape problem recognition. We formalize them later.

The other weights shape the DM's sensitivity to her tastes u and to information about e . The perceived value of hedonic feature $i \in \Phi_H$ is, given weight $\alpha_{P,i}$, equal to:

$$(1) \quad u_i(\alpha_P) = \alpha_{P,i} \cdot u_i + (1 - \alpha_{P,i}) \cdot \bar{u}_i,$$

where u_i is the value of the hedonic in the atom and $\bar{u}_i$ is its average value across atoms in O .⁴ For instance, less than full attention to the jam's price, $\alpha_{P,Mj} < 1$, $j \in \{g, b\}$, reduces perceived price variation in the choice set.

2. Formally, in atom $y = (u, e)$ let $e_i \subseteq \Omega$ be a generic entry of e , $i \in \Phi_E$. Payoff u is delivered in event $\cap_{i \in \Phi_E} e_i$, whose probability is $\mathbb{P}(e) = \mathbb{P}(\cap_{i \in \Phi_E} e_i)$.

3. The atoms are $(q_g, \eta m_g, 0, 0, g)$ and $(0, 0, q_b, \eta m_b, b)$. The spoiled jam is not delivered in state g so its value in this state is zero, and similarly for other non-delivered features.

4. Formally, $\bar{u}_i = \frac{\sum_{o \in O} \sum_{(u,e) \in o} u_i}{\sum_{o \in O} |o|}$.

With a single event feature W , the perceived probability of its realization e is, given weight $\alpha_{P,W}$, equal to:

$$(2) \quad \Pr(e; \alpha_P) = \frac{\mathbb{P}(e)^{\alpha_{P,W}}}{\sum_{e' \in W} \mathbb{P}(e')^{\alpha_{P,W}}}.$$

Less than full attention, $\alpha_{P,W} < 1$, causes the DM to smear the probability distribution of W toward a uniform, fully so if $\alpha_{P,W} = 0$. The DM overestimates the event that the jam is spoiled if she does not attend to its small numerical probability.

The value of atom y under representation α_P is the product of its perceived probability in [equation \(2\)](#) and its total perceived payoff computed using [equation \(1\)](#):

$$(3) \quad v(y; \alpha_P) = \Pr(e; \alpha_P) \sum_{i \in \Phi_H} u_i(\alpha_P),$$

and the value of lottery o is the sum of its atomic values $v(o; \alpha_P) = \sum_{y \in o} v(y; \alpha_P)$.

In our example, up to a choice-irrelevant affine transformation, valuation satisfies:

$$(4) \quad v(o; \alpha_P) = \sum_{e \in \{g, b\}} \mathbb{P}(e)^{\alpha_{P,W}} [\alpha_{P,Qe} (q_e - \bar{q}_e) + \alpha_{P,Me} \eta(m_e - \bar{m}_e)].$$

If attention is full, $\alpha_P = 1$, we recover expected utility. If attention is not full, [equation \(4\)](#) nests well-known behavioral models.

If the jam is always good ($\mathbb{P}(b) = 0$) and if $m_g = -p$, we have riskless choice with weighted utility, $v(o; \alpha_P) = \alpha_{P,Q} (q - \bar{q}) - \alpha_{P,M} \eta(p - \bar{p})$. Attention to money relative to quality $\frac{\alpha_{P,M}}{\alpha_{P,Q}}$ pins down price sensitivity, as in [Bordalo, Gennaioli, and Shleifer \(2013\)](#). Critically, if the DM neglects the spoiled jam payoffs ($\alpha_{P,Qb} = \alpha_{P,Mb} = 0$), she chooses “as if” choice is riskless even if $\mathbb{P}(b) > 0$.

In monetary lotteries ($q_e = 0$ for all $e \in \{g, b\}$), inattention to the data-generating process (DGP), $\alpha_{P,W} < 1$, gives overweighting of unlikely events ([Kahneman and Tversky 1979](#)). Higher attention to the upside versus downside monetary payoff $\frac{\alpha_{P,Mg}}{\alpha_{P,Mb}}$ yields payoff-driven risk seeking ([Bordalo, Gennaioli, and Shleifer 2012](#)).

This setting also embeds statistical problems: a hypothesis is a lottery that delivers a payoff of \$1 if an outcome of W occurs and zero otherwise. We later allow for multidimensional e , which produces long-standing judgment biases.

Crucially, attention weights to features of tastes and information, $(\alpha_{P,i})_{i \in \Phi_H \cup \Phi_E}$, depend on attention to context, given by weights

$(\alpha_{P,i})_{i \in \Phi_K}$. The reason is that attention to context shapes problem recognition, as we show next.

II.B. Recognition and Choice

The context vector κ_P reports standard relevant features of P , such as the choice set, the range of hedonics (e.g., available qualities or prices), and the list of events.⁵ It also reports irrelevant features such as time, location, or packaging, which can affect recognition. If the jam is offered with rich sensory features (e.g., a fancy jar) and the DM attends to them, she represents the problem P as “consuming” rather than “buying.” This prompts focus on jam qualities and neglect of prices or risk. If the DM attends to the supermarket location, she recognizes P as “buying,” causing focus on price and neglect of quality.

Attention to context and recognition depend on the categories of problems the DM solved in the past. To see how, we summarize the current problem as a vector of attention plus context (α_P, κ_P) .

1. Categories. The DM’s memory database is partitioned into a set of categories C . Past problems in category $c \in C$ are also summarized by an attention plus context “prototype” vector (α_c, κ_c) and by their time-discounted frequency $F_c \in \mathbb{R}_+$.⁶

Attention α_c is the prototypical representation used in category c problems. It specifies the context diagnostic of experiences in c (Rosch and Lloyd 1978) via binary attention $\alpha_{c,i} \in \{0, 1\}$ for $i \in \Phi_K$. Eating jam mostly occurs at home, but in a wide range of weather conditions, so location feature i is diagnostic for this category, $\alpha_{c,i} = 1$, but weather feature i' is not, $\alpha_{c,i'} = 0$. Vector α_c also reports the attention $\alpha_{c,i}$ the DM paid to hedonics and events in c , $i \in \Phi_H \cup \Phi_E$. When eating jam, the DM is more focused on quality and less on price than when buying jam. Experiences “glue” context with attention to relevant features.

5. Features in κ_P can also include the average price level (expensive versus cheap goods problem) or the average probabilities of specific events (high versus low risk problem).

6. For a DM facing a problem in period $t \in \mathbb{N}$, a category $c \subseteq \{1, \dots, t-1\}$ collects a set of past periods. Recency-weighted frequency is $F_c = \sum_{\tau \in c} \delta^{t-\tau}$, $\delta \in (0, 1)$. The “prototype” can be interpreted as the average attention $\alpha_c = \frac{1}{|c|} \sum_{\tau \in c} \alpha_\tau$ used in these problems and the “average” context κ_c where, for every $i \in \Phi_K$, $\kappa_{c,i}$ minimizes some discrepancy from $(\kappa_{\tau,i})_{\tau \in c}$.

2. *Recognition.* To solve P , the DM must decide: “What kind of problem is this? What matters for solving it?” To this end, she chooses attention α_P and a category c to maximize the total similarity between the current problem (α_P, κ_P) and a category prototype (α_c, κ_c) . Similarity is measured by a function $S[(\alpha_P, \kappa_P), (\alpha_c, \kappa_c)]$ that decreases if (i) the context κ_c of past problems in c is different from current context κ_P , or (ii) the representation α_c used for problems in c differs from the current attention vector α_P . We use the separable form:

$$(5) \quad S[(\alpha_P, \kappa_P), (\alpha_c, \kappa_c)] = 1 - \frac{\sum_{i \in \Phi} d(|\alpha_{P,i} - \alpha_{c,i}|) + \sum_{i \in \Phi_K} \alpha_{P,i} \alpha_{c,i} d_i(\kappa_{P,i}, \kappa_{c,i})}{|\Phi| + |\Phi_K|}.$$

Discrepancies in attention are measured by $d : [0, 1] \rightarrow [0, 1]$, which is strictly increasing and convex. Discrepancies in context feature $i \in \Phi_K$ are measured by a distance d_i with values in $[0, 1]$. This functional form helps tractability, but our results rely on the general idea—central in psychology (Nosofsky 1986)—that similarity falls in differences, and more so when these are more attended to.⁷

Total similarity multiplies S by F_c , capturing the idea that more familiar categories, higher F_c , are easier to retrieve. Similarity is also perturbed by a type I extreme-value shock ϵ_c with precision parameter $\lambda > 0$, reflecting random attention to categories. This error structure is not necessary but yields convenient closed forms.

3. *Choice.* Conditional on a draw $(\epsilon_c)_{c \in C}$, the DM solves P as follows:

$$(6) \quad (\alpha_P^*, c^*) = \operatorname{argmax}_{\alpha_P \in [0,1]^\Phi, c \in C} F_c S[(\alpha_P, \kappa_P), (\alpha_c, \kappa_c)] + \epsilon_c,$$

$$(7) \quad o^* = \operatorname{argmax}_{o \in O} v(o; \alpha_P^*).$$

In equation (6) the DM represents the problem using category c^* . She then uses the endogenous weights α_P^* to value and pick the best lottery o^* in equation (7). In rational choice, attention is full $\alpha_P^* = \mathbf{1}$ and the DM maximizes $v(o; \mathbf{1})$. In our model,

7. Consistent with an attention-dependent metric, Goldstone (1994) shows that people trained to relate specific features among objects in one task subsequently perceive differences across those features to be larger, and differences across other features to be smaller.

keeping $v(o; \mathbf{1})$ fixed, attention and choice are heterogeneous due to experiences F_c , and unstable due to context κ_P .

Rational accounts of heterogeneity and instability assume that hedonics in $v(o; \mathbf{1})$ directly depend on experiences through habits, as in [Stigler and Becker \(1977\)](#), or on context such as advertising, as in [Becker and Murphy \(1993\)](#). This account is tautologically consistent with choice because tastes are not measured directly, only choices are. A deeper understanding of choice can be achieved by measuring beliefs, the probabilities in $v(o; \mathbf{1})$, for instance, about returns to human capital ([Manski 1995](#)) or financial assets ([Bordalo et al. 2024, 2025c](#)). Measured beliefs narrow down unobserved variation in tastes. Our model highlights a deeper force: shifting attention to features. The way the DM thinks about a problem can affect her weighting of hedonics and events, which may be confused with differences in tastes and information, respectively. Structure on α_P is needed to identify the drivers of choice.

[Equation \(6\)](#) imposes such a structure and links the theory's endogenous objects, attention and choice, to proxies for the cognitive process: experiences, features of context, and similarity. A growing body of work measures these proxies in the field (e.g., using administrative and text data), in the lab, or in surveys, and shows that even if they are unrelated to choice-relevant tastes or information, they drive behavior through selective attention or memory mechanisms ([Ba, Bohren, and Imas 2024](#); [Cenzon 2025](#); [Taubinsky et al. forthcoming](#); [Fan, Liang, and Peng forthcoming](#)). For consumer choice, [Bordalo et al. \(2025b\)](#) show that advertising of a brand increases consumption of other brands with similar ads, especially for consumers who have more experience with those brands. In belief formation, [Bordalo et al. \(2026b\)](#) show that macroeconomic narratives and the entailed expectations are influenced by recall of uninformative personal experiences, especially in contextually similar domains.

Our model shows how to use these proxies to study selective top-down attention driven by problem recognition, which shapes choice even if domain-relevant tastes or information are fully available to the DM, such as when choosing at a supermarket between two known brands or when evaluating two statistical hypotheses based on a described DGP. The problem recognition step makes further predictions on the reasoning and narratives used to justify choices, which are now routinely measured in surveys

(Andre et al. 2021; Haaland et al. 2024) and on how these depend on bottom-up salient features (Li and Camerer 2022).

Next we illustrate our model's general predictions on representations, and link these predictions to choices and judgments.

II.C. General Predictions on Representations

Consider how the DM sets optimal attention α_P to match problem P to a given category c . By equation (6), an interior solution for a generic feature $i \in \Phi_K$ satisfies the first-order condition:

$$(8) \quad d'(\alpha_{c,i} - \alpha_{P,i}) = d_i(\kappa_{P,i}, \kappa_{c,i}) \alpha_{c,i}.$$

Setting a representation α_P close to the categorical one α_c increases similarity to c (the left-side term), but may highlight differences in context diagnostic of the category, $d_i(\kappa_{P,i}, \kappa_{c,i}) \alpha_{c,i} > 0$ on the right side. We then obtain:

PROPOSITION 1. Optimal attention at fixed category c ,

$$\alpha_P(c) = \operatorname{argmax}_{\alpha_P \in [0,1]^\Phi} S[(\alpha_P, \kappa_P), (\alpha_c, \kappa_c)],$$

is equal to category attention, $\alpha_{P,i}(c) = \alpha_{c,i}$, unless feature $i \in \Phi_K$ is diagnostic of c and discrepant with P , $d_i(\kappa_{P,i}, \kappa_{c,i}) \cdot \alpha_{c,i} > 0$. In the latter case, attention to that feature is shrunk toward zero, $\alpha_{P,i}(c) \leq \alpha_{c,i}$, and the more so the higher is $d_i(\kappa_{P,i}, \kappa_{c,i})$.⁸

For hedonics and events, attention fully adapts to the category's α_c . The choice implication is that the sensitivity of the DM's valuation to price, qualities, and so on is shaped by the category.

Attention is instead lower than in α_c for contextual features where the problem differs from the category, the more so the greater the difference. As a consequence, the perceived similarity of problem P to a category c is endogenous, given by $S(P, c) = F_c[1 - d(P, c)]$, where

$$d(P, c) = \min_{\alpha_P \in [0,1]^\Phi} \frac{\sum_{i \in \Phi} d(|\alpha_{c,i} - \alpha_{P,i}|) + \sum_{i \in \Phi_K} \alpha_{P,i} \alpha_{c,i} d_i(\kappa_{P,i}, \kappa_{c,i})}{|\Phi| + |\Phi_K|}$$

is the minimized distance from c . Endogenous inattention reduces the sensitivity of similarity $S(P, c)$ to contextual discrepancies $d_i(\kappa_{P,i}, \kappa_{c,i})$, with important implications. Equilibrium representations work as follows.

8. Here optimal attention is unique. The Online Appendix relaxes this assumption.

PROPOSITION 2. Suppose that $\alpha_P(c) \neq \alpha_P(c')$ for all $c \neq c'$. The DM's representation satisfies

$$(9) \quad \Pr(\alpha_P^* = \alpha_P(c)) = \frac{\exp\{\lambda F_c[1 - d(P, c)]\}}{\sum_{c' \in C} \exp\{\lambda F_{c'}[1 - d(P, c')]\}} \quad \forall c \in C.$$

Problem P is more likely to be represented using $\alpha_P(c)$ when category c :

- i. has a context closer to the current one, $\frac{\partial \Pr(\alpha_P^* = \alpha_P(c))}{\partial d_i(\kappa_{P,i}, \kappa_{c,i})} \leq 0$.
- ii. was used more frequently or recently, $\frac{\partial \Pr(\alpha_P^* = \alpha_P(c))}{\partial F_c} \geq 0$.

Due to the shock ϵ_c categorization is stochastic, causing discrete switches in attention to options' attributes. Our first behavioral prediction follows: within-person multimodal valuation. In similar inference problems, a person can switch from using only the base rate to using only the likelihood, as shown in [Bordalo et al. \(2026a\)](#). This pattern does not arise with noisy perception of payoffs/probabilities ([Woodford 2012](#)) or of the optimal action ([Enke and Graeber 2023](#)). In these models, the weighting of the features is stable, so valuation is unimodal. In the data it is not, and our model explains why.

Relatedly, context creates systematic instability: a person is less likely to use a category c when a current context feature $\kappa_{P,i}$ becomes more discrepant from the category's $\kappa_{c,i}$. A second behavioral prediction follows: valuation predictably switches from one mode to another after changes in context. Changing the inference context from a balls and urns problem ([Edwards 1968](#)) to a normatively identical taxicabs problem ([Kahneman and Tversky 1972](#)) causes many people to switch from the base rate to the likelihood even though statistics are unchanged ([Bordalo et al. 2026a](#)).

Finally, experiences create systematic heterogeneity: people more familiar with a category c (higher F_c) are more likely to use it. A third behavioral prediction follows: valuation differs across people, due to not only tastes but also differential attention to features. In inference problems, some people are more likely than others to use the base rate or the likelihood, despite common incentives and information ([Bordalo et al. 2026a](#)). This heterogeneity reflects the fact that people endogenously neglect different features of context (those that are dissonant with the categories they are familiar with). It also implies that the overall distribution of valuations may appear unimodal (if many categories are possible) even if individual distributions are not.

Our theory generates famous problem-solving errors (Bassok 2003). First, training on a problem improves performance but expertise is too context specific (Chase and Simon 1973; Camerer and Johnson 1991; Green, Rao, and Rothschild 2019). A DM trained on inference with numerical priors and likelihoods can reach the correct solution with arbitrary values and yet fail to apply Bayes rules or conditioning when the problem—and hence its context—has no numbers (such as in the Linda problem in Kahneman and Tversky 1983). Second, experience with a problem (high F_c) causes overgeneralization: some people’s familiarity with solving inference problems may cause them to focus on the empirical share of good to bad signals even when this statistic is irrelevant, as in the case of the gambler’s fallacy studied in Section IV.

Next we derive the implications of our model for riskless choice, statistical problems, and risky choice. Specific predictions depend on the set of categories C , which we take as given, as does Mullainathan (2002). To discipline our analysis we restrict attention to four categories, two in riskless choice, two in statistical problems. The conclusions discuss ways to endogenize and measure a broader set of categories.

III. REPRESENTATIONS IN RISKLESS CHOICE

The DM must choose whether or not to buy a jam. The state is not yet realized, so we are in the case of equation (4). The modal state g exhibits “normal” quality $q_g = q$ and price $p_g = p$. State b is an unlikely “shock” that the DM has never experienced and that throughout this section will be assumed to have subjective probability $\mathbb{P}(b) = 0$, so that its quality q_b and price p_b do not play a role.⁹ Context is a unidimensional situation, feature K , with value $\kappa_P \in [0, 1]$. There are two categories: “buying” and “consuming.”

i. Buying. The buying category, denoted *buy*, reflects experiences in which the DM focuses on the pain of paying for jam and partly on its forecasted quality (otherwise she would not buy), so $\alpha_{Mg} = 1 > \alpha_{Qg} = \alpha$. The unlikely shock b is neglected, $\alpha_{Qb} = \alpha_{Mb} = 0$. Diagnostic context embeds various features such as being at the supermarket,

9. This will be relaxed in Section V.B.

captured by $\kappa_{buy} = 0$. The *buy* prototype $(\alpha_{buy}, \kappa_{buy})$ is thus the vector $(\alpha_{buy, Qg}, \alpha_{buy, Mg}, \alpha_{buy, Qb}, \alpha_{buy, Mb}, \alpha_{buy, W}, \alpha_{buy, K}, \kappa_{buy}) = (\alpha, 1, 0, 0, \alpha_{buy, W}, 1, 0)$.

ii. Consuming. The consuming category, denoted *con*, reflects experiences in which the DM focuses on the pleasure of jam, while paying less attention to the purchase price. Ex post, the DM focuses on the realized quality during consumption; ex ante she focuses on the modal quality. From the ex ante perspective, which is relevant for our current exercise, the DM attends to quality and partially to price, $\alpha_{Qg} = \alpha_{Qb} = 1 > \alpha = \alpha_{Mg} = \alpha_{Mb}$. Diagnostic context reflects features of tasting, such as being in one's kitchen, captured by $\kappa_{con} = 1$. The consuming prototype is then $(\alpha_{con, Qg}, \alpha_{con, Mg}, \alpha_{con, Qb}, \alpha_{con, Mb}, \alpha_{c, W}, \alpha_{con, K}, \kappa_{con}) = (1, \alpha, 1, \alpha, \alpha_{con, W}, 1, 1)$.

In this setup, the experiences in *con* entail an accurate perception of normal quality q , whereas experiences in *buy* entail an accurate perception of normal price p and choice is seen as riskless. To properly trade off q and p the DM should integrate the valuation of quality in *con* and of price in *buy*. However, *con* and *buy* are segregated in different contexts κ_c and they compete for the DM's representation based on the DM's familiarity F_c and the current context κ_P .

III.A. The Context-Dependent Consumer

With attention distance $d(x) = \frac{x}{2}$, and context distance $d_Z(z, z') = |z - z'|$, where Z denotes the single context dimension in κ_P , such as location, we have:

(10)

$$S[(\alpha_P, \kappa_P), (\alpha_c, \kappa_c)] = 1 - \frac{[|\alpha_{P, Qg} - \alpha_{c, Qg}| + |\alpha_{P, Mg} - \alpha_{c, Mg}| + |\alpha_{P, Z} - 1| + 2\alpha_{P, Z} \cdot |\kappa_P - \kappa_c|]}{8}.$$

For each category $c \in \{con, buy\}$, the DM sets the attention $\alpha_P(c)$ that maximizes similarity, producing two competing representations. The resulting similarities to P give, as a function of current context κ_P , the following result.

PROPOSITION 3 (REPRESENTATION AND SIMILARITY). Given optimal weights $\alpha_P(c) = (\alpha_{Qg}(c), \alpha_{Mg}(c), \alpha_Z(c))$ on normal quality, price, and on context for category c , we have:

- i. To represent P as *buy*, the DM sets $\alpha_P(\text{buy}) = (\alpha, 1, 1)$ if $\kappa_P < \frac{1}{2}$ and $\alpha_P(\text{buy}) = (\alpha, 1, 0)$ if $\kappa_P > \frac{1}{2}$. Thus, $S(P, \text{buy}) = F_{\text{buy}} \max \left[1 - \frac{\kappa_P}{4}, \frac{7}{8} \right]$.
- ii. To represent P as *con*, the DM sets $\alpha_P(\text{con}) = (1, \alpha, 1)$ if $\kappa_P > \frac{1}{2}$ and $\alpha_P(\text{con}) = (1, \alpha, 0)$ if $\kappa_P < \frac{1}{2}$. Thus, $S(P, \text{con}) = F_{\text{con}} \max \left[\frac{3+\kappa_P}{4}, \frac{7}{8} \right]$.

The DM focuses on the jam's price and neglects its quality when using *buy*, and the reverse with *con*. Moreover, she focuses on context when it resembles the category, namely, low κ_P for *buy* and high for *con*, and neglects it otherwise. Neglecting the problem's context reduces similarity to the category (the DM knows that context should match), but focusing on the associated discrepancy reduces similarity even more.

Maximal similarity $S(P, c)$ increases in the category's frequency F_c and decreases in the distance between its context κ_c and κ_P . With negligible shocks ϵ_c , $\lambda \rightarrow \infty$, recognition in [equation \(6\)](#) yields the following result.

PROPOSITION 4 (CATEGORIZATION AND VALUATION). The DM categorizes P as *buy* if $\kappa_P < \kappa \left(\frac{F_{\text{buy}}}{F_{\text{con}}} \right)$ and as *con* otherwise, where $\kappa(\cdot)$ is increasing. Valuation satisfies:

$$(11) \quad v(\text{jam}; \alpha_P^*) = \begin{cases} \alpha q - \eta p & \text{if } \kappa_P < \kappa \left(\frac{F_{\text{buy}}}{F_{\text{con}}} \right) \\ q - \alpha \eta p & \text{if } \kappa_P \geq \kappa \left(\frac{F_{\text{buy}}}{F_{\text{con}}} \right) \end{cases}.$$

The DM recognizes P as “buying” if context κ_P is sufficiently similar to $\kappa_{\text{buy}} = 0$, and as consuming otherwise. *buy* more likely prevails if this category is relatively more frequent, higher $\frac{F_{\text{buy}}}{F_{\text{con}}}$. [Figure I](#) assumes that the categories are equally frequent $F_{\text{buy}} = F_{\text{con}} = F$ and plots the similarity of P to each category in Panel A and the valuation function (11) in Panel B. Two properties stand out. First, valuation of the jam is unstable. Categorization changes from *buy* to *con* as context crosses $\kappa_P = \frac{1}{2}$ from below, causing attention to quality and valuation to shift upward. The DM is “stingy at the supermarket” ($\kappa_P < \frac{1}{2}$) and may fail to buy even if $q > \eta p$, but “neglects opportunity cost at home” ($\kappa_P > \frac{1}{2}$), enjoying the jam even if $q < \eta p$. Second, price sensitivity is high for $\kappa_P < \frac{1}{2}$ and low for $\kappa_P > \frac{1}{2}$. A vertically superior jam located on a rational indifference curve $q - \eta p = u$ is undervalued in *buy* and overvalued in *con*.

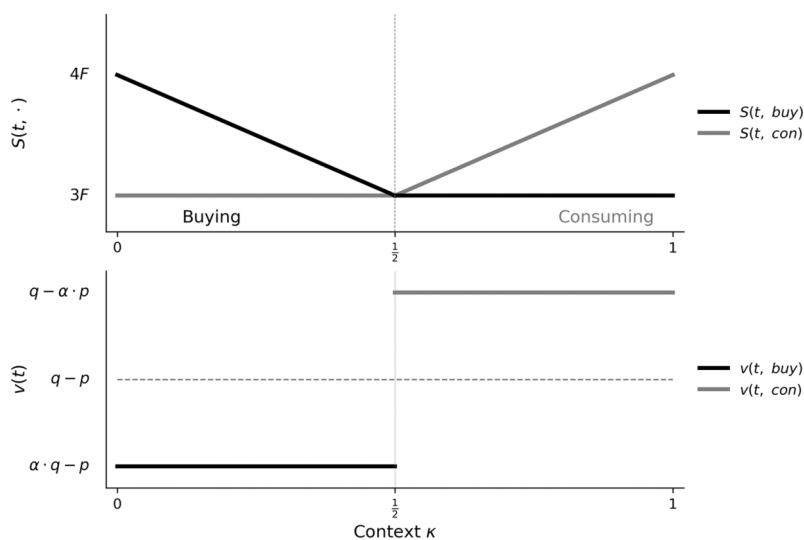


FIGURE I

Context, Representations, and Consumer Choice

Panel A: Similarity $S(P, c)$ to categories *con* and *buy*, equation (10). Panel B: Valuation, equation (11), which follows from categorization into the most similar category.

1. First Model Prediction. A person's price sensitivity should weakly decline if context becomes more similar to consuming experiences and less so to buying ones. Selling the jam in a fancy jar on a wooden tasting board evokes special occasions of enjoying treats at home, while reducing similarity to buying staple products, boosting sensitivity to quality and reducing that to price.

This prediction accounts for a realistic instability in price elasticity. Wakefield and Inman (2003) show that a person is price inelastic if she categorizes the good as "hedonic" and price elastic if she categorizes the good as "functional." This phenomenon reflects the role of context, embedded in the good itself or in the situations in which it is typically consumed. The context of a "Friday night takeout" prompts focus on pleasure due to past treat experiences, encouraging people to order an extra appetizer or dessert. The context of a work lunch prompts focus on minimizing spending, increasing the price sensitivity.

Context-based instability explains why uninformative advertising and spurious differentiation can transform a good from “functional” to “hedonic”: they increase the similarity of context to *con*, leading to the relative neglect of price. By creating a “living room” experience, a comfortable café draws attention to coffee taste and sociality, increasing valuation even for people who get the drink to go. By drawing attention to sources in tropical islands or glaciers, branded water draws attention to its freshness, increasing valuation even for a standard product. A context-driven shift from “buying” to “consuming” causes decommodification.¹⁰

The model also makes predictions based on experiences. Suppose that the frequency of *buy* is higher than that of *con*, $F_{buy} > F_{con}$. In Figure II the similarity to *buy* uniformly increases from the dashed to the solid black line, promoting that category at any context κ_P . The consumer is more price elastic even if her true marginal utility of money η is the same. Endogenous attention causes insensitivity to discrepancies with *buy*, leading to a much higher threshold κ^* , also relative to the cutoff κ^{**} attained if the DM fully attends to context (dash-dotted black line in Figure II).

2. Second Model Prediction. Take two people, A and B, having the same income and tastes. The person with the higher relative frequency of buy experiences (higher $\frac{F_{buy}}{F_{con}}$) should be weakly more price elastic in any context κ_P . Familiarity with reasoning about prices prompts the DM to recognize the current choice as “buying,” increasing the price sensitivity.

The role of experiences explains the puzzling heterogeneity of price elasticities. People with experiences of poverty are more price elastic across situations (Shah et al. 2018). Having experienced high opportunity costs, they tend to see price as highly relevant. This is overgeneralization, not a higher opportunity cost η or a Bayesian prior. It can cause large mistakes when quality q is high, such as saving on valuable medical treatment by focusing on low co-pays (Baicker, Mullainathan, and Schwartzstein 2015; Chandra, Flack, and Obermeyer 2024). Price focus can persist even if the DM is no longer poor, contributing to variation

10. Bordalo, Gennaioli, and Shleifer (2016) first explored decommodification, and its inverse, commoditization, in terms of selective attention to quality and prices, respectively. The mechanism there is bottom-up contrast in a vertically differentiated market. In Section V we examine how bottom-up drivers of salience complement the categorization mechanism.

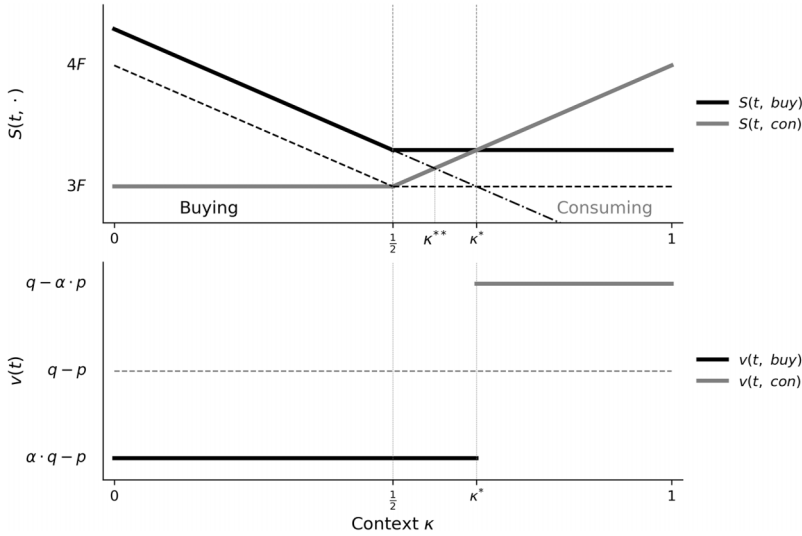


FIGURE II

Frequent Categories Promote Neglect of Discrepant Context

Increasing the frequency F_{buy} increases similarity $S(P, buy)$, increasing the range of context κ_P where P is categorized as *buy*. This includes the range $\kappa_P \in (\frac{1}{2}, k^*)$ where the DM maximizes similarity by neglecting discrepant context.

in price elasticities and thriftiness for given income and wealth (Hoch et al. 1995; Rick, Cryder, and Loewenstein 2008). Measured experiences help test our mechanism: Link et al. (2025) show that people with high inflation experiences attend more to inflation today.¹¹

III.B. Mental Accounting

People use different accounts to track costs and benefits in different situations, leading to opportunity cost neglect, sunk cost fallacy, nonfungibility of money, and so on. Consider the following examples.

11. Context-based instability and experience-based heterogeneity interact in driving categorization. Instability can explain why even the poor can “splurge” in contexts associated with consumption pleasure, such as festivals or “treat” goods such as cigarettes (Banerjee and Duflo 2007), or why a person who grew up poor is less price elastic in new goods, such as iPhones or travel, whose context is distant from her *buy* experiences in items such as clothes or food.

- i. *Opportunity Cost Neglect*. Many years ago a person bought for \$20 a bottle of wine worth \$75 today. The person drinks the wine today. What is the cost she feels? Many people say zero or \$20 (Thaler 1980). They neglect the opportunity cost of drinking, the \$75 market price.
- ii. *Sunk Cost Fallacy*. A person bought a \$20 ticket to a football game that takes place a month later. On the game day, there is a severe blizzard. Does the person drive to the game? Would she drive had she been given the ticket for free? Frequent answers are “yes” to the first question, and “no” to the second, violating revealed preference: if the blizzard is severe enough, not driving to the game is optimal regardless of having paid the ticket price.

Opportunity cost neglect has been attributed to the temporal remoteness of the wine’s purchase price (Gourville and Soman 1998), while the sunk cost fallacy has been explained with diminishing sensitivity (Thaler 1980) or distaste for “waste” (Shafir and Thaler 2006).¹²

We show that both phenomena arise from categorization in our baseline consumer problem, which yields new predictions. In both examples, we are in an ex post situation where the “not-normal” state b has realized. In the wine problem, a shock affects the price, which increases by $\Delta p > 0$. In the football problem, a shock affects quality, which changes by $\Delta q < 0$. Anomalies then arise because the consumer neglects these shocks due to categorization into consuming or buying.

In the wine example, the DM must evaluate the cost of drinking so both the original purchase price p and the capital gain Δp are relevant. In the football example, the DM must evaluate the benefit of driving to the game, so the quality forecast q at the time of buying and the disutility of driving Δq are relevant. As a function of attention weights, valuations in the two problems are:

$$(12) \quad v(\text{cost of drinking}; \alpha_P) = \alpha_{P,M}p + \alpha_{P,\Delta M}\Delta p,$$

12. Kőszegi and Matějka (2020) offer a model of rational inattention that produces instances of behavior linked to mental accounting when hedonics are noisy. For substitute goods, attention is optimally focused on features that distinguish them, leading to category budgets. The same mechanism leads to naive diversification if goods are complements. However, this model does not explain the sunk cost fallacy or the wine example, or more generally the role of context in driving instability.

$$(13) \quad v(\text{driving}; \alpha_P) - v(\text{not driving}; \alpha_P) = \alpha_{P,Q}q + \alpha_{P,\Delta Q}\Delta q.$$

With full attention, decisions are rational. In the wine problem, with full attention to monetary payoffs $\alpha_{P,M} = \alpha_{P,\Delta M} = 1$ valuation equals the market price, $p + \Delta p = \$75$ (equation (12)). In the football problem, $\alpha_{P,Q} = \alpha_{P,\Delta Q} = 1$ and the decision is to go to the game if and only if $q + \Delta q > 0$.

However, the consuming and buying categories do not attend to these features fully. *Con* fully attends to actual quality, which is now realized in b , so $\alpha_{con,Q} = \alpha_{con,\Delta Q} = 1$, but it only partially attends to the price at which the good was bought, $\alpha_{con,M} = \alpha \in (0, 1)$, and it fully neglects the capital gain, which is not typically attended to when enjoying consumption, $\alpha_{con,\Delta M} = 0$. On the other hand, *buy* fully attends to the price paid and partially attends to the quality forecast ex ante, during the buying decision, $\alpha_{buy,M} = 1$ and $\alpha_{buy,Q} = \alpha \in (0, 1)$. It neglects both quality shocks and capital gains, $\alpha_{buy,\Delta Q} = \alpha_{buy,\Delta M} = 0$, which were not attended to when buying. These categories prompt evaluations:

$$(14) \quad v(\text{cost of drinking}; \alpha_P) = \begin{cases} \alpha p & \text{if } \alpha_P = \alpha_P(\text{con}) \\ p & \text{if } \alpha_P = \alpha_P(\text{buy}) \end{cases},$$

$$v(\text{driving}; \alpha_P) - v(\text{not driving}; \alpha_P) = \begin{cases} (q + \Delta q) & \text{if } \alpha_P = \alpha_P(\text{con}) \\ \alpha q & \text{if } \alpha_P = \alpha_P(\text{buy}) \end{cases}.$$

In the wine problem, “consuming” focuses the DM on the pleasure of drinking, causing opportunity cost neglect. If $\alpha = 0$ the DM reports a zero cost, as do many experimental subjects.¹³ “Buying” instead focuses the DM on the purchase price p , which is also reported by several subjects. The capital gain is still neglected: it was not attended to when buying the bottle.

In the football problem, “consuming” focuses the DM on the qualities of driving: seeing the game q but also on the blizzard $\Delta q < 0$, leading to the rational evaluation. “Buying” instead focuses the DM on the purchase price p and the normal benefit of seeing the game q . The blizzard shock is neglected: it was not attended to when buying the ticket.

The context κ_P of the vignette affects the modal answer by shaping similarity to different categories. In the wine problem, the description “drinks the wine” prompts similarity to *con* (high

13. During consumption, with prices not explicitly mentioned, we often feel no opportunity cost. Frederick et al. (2009) show that describing how savings can be used for other purchases substantially decreases the probability of spending.

κ_P in Figure I), favoring opportunity cost neglect. In the football problem, the choice of “going versus not going to the game” evokes a buying problem (low κ_P in Figure I), promoting neglect of the blizzard, but less so if the ticket was given for free and did not involve a decision to buy.

This analysis then yields two comparative statics:

- i. Frequency. People with higher F_{buy} exhibit less opportunity cost neglect and more sunk cost fallacy. These could be people with poverty experiences or, in the wine example, people who have bought wine but have not yet drunk it (so recency-weighted frequency of buying is high). In the football example, people who bought season tickets face one buying experience but many consuming ones; they have a high F_{con} , which undermines *buy* and reduces the sunk cost fallacy.
- ii. Instability. Making the blizzard more salient in the description or making an ex ante plan for bad weather should increase reliance on *con*, reducing the sunk cost fallacy. Making the reselling of the wine more salient, either as a forgone opportunity with consumption or as a possibility when buying, should hinder *con* and opportunity cost neglect. A wine trader has a frequent *buy* category that attends to the capital gain, so she exhibits less opportunity cost neglect (List, 2004).

These predictions are based on the same mechanism: context and experiences prompt different categorizations of the same problem, focusing attention on different features and shaping choice. Problem recognition also explains nonfungibility: transferring money into a category prompts recognition of “buying in that category,” promoting in-category spending. Mental categories also create commitment (Thaler 1980): setting up a “rainy day” account prompts recognition of “minimizing opportunity costs,” raising similarity to *buy*, and discouraging spending. Account names focus the consumer on different goals, affecting how they recognize the problem and make their choices (Henderson and Peterson 1992).

IV. REPRESENTATIONS IN JUDGMENTS

Expectations and probability judgments highlight the role of selective attention to different pieces of information. People with

high-inflation experiences attend more to inflation and its surges than people without such experiences (Gennaioli et al. 2024; Link et al. 2025). Professionals and households use narratives or models that attend to different data and change over the cycle (Andre et al. 2021; Bastianello, Décaire, and Guenzel 2024). Bordalo et al. (2026a) show that famous biases in i.i.d. draws and inference problems (Benjamin 2019) are also linked to changing attention to specific features.

These findings reject models of optimal beliefs or stable heuristics/distortions of true probabilities (Grether 1980). We show that this heterogeneity and instability emerges naturally from endogenous recognition of different statistical categories, unifying judgment and decision-making, and yielding new predictions.

IV.A. Generalizing the Model

To allow for general DGPs, the event vector e is multidimensional. In balls and urns problems, one feature is urn selection $e_U \in \{\text{urn } A, \text{urn } B\}$, the other is the color of the ball drawn from the urn $e_B \in \{\text{red}, \text{blue}\}$. The event of selecting an urn and then drawing a ball from it is $e = (e_U, e_B)$, with probability $\mathbb{P}(e) = \mathbb{P}(e_B | e_U) \cdot \mathbb{P}(e_U)$. We generalize to multistep causal models by an order $<$ over event features in Φ_E . The true probability of atom $y = (u, e)$ with $e = (e_s)_{s \in \Phi_E}$ is:

$$(15) \quad \mathbb{P}(e) = \prod_{s \in \Phi_E} \mathbb{P}(e_s | \cap_{j < s} e_j).$$

Attention shapes the DM's sensitivity to the probability of different events, as in equation (2). Limited attention to an event s , $\alpha_{P,s} < 1$, distorts its conditional probability into $\mathbb{P}(e_s | \cap_{j < s} e_j)^{\alpha_{P,s}}$. In the limit of zero attention, $\alpha_{P,s} = 0$, event s is “edited out,” which prevents the conditioning of downstream events $s' > s$ on it. Formally, let edited vector $e(\alpha_P)$ report the correct event, $e_s(\alpha_P) = e_s$, if $\alpha_{P,s} > 0$ and set $e_s(\alpha_P) = \Omega$ otherwise, so that any realization of a neglected event is allowed. The perceived probability equation (15) is then:

$$(16) \quad \Pr(e; \alpha_P) \propto \prod_{s \in \Phi_E} \mathbb{P}(e_s | \cap_{j < s} e_j(\alpha_P))^{\alpha_{P,s}},$$

normalized so that probabilities across all e add to one. Equation (16) nests Grether's (1980) formula but with endogenous weights. Valuation works as in Section II: atom y is replaced by its edited

version $y(\alpha_P) = (u(\alpha_P), e(\alpha_P))$ where $u(\alpha_P)$ satisfies [equation \(1\)](#). As in [equation \(3\)](#), the value of $y(\alpha_P)$ multiplies its perceived probability and hedonics. The value $v(o; \alpha_P)$ of the edited lottery adds the values of its atoms. Full attention, $\alpha_P = \mathbf{1}$, recovers rationality.¹⁴

IV.B. Biases in i.i.d. Draws and Inference

To jointly capture judgments about inference and i.i.d. sequences, consider a problem with $N + 2$ event features. First, a coin (or urn) is selected of type $e_U \in \Theta$. This is followed by a sequence of $N \in [1, \bar{N}]$ i.i.d. draws $(e_j)_{j \in \{1, \dots, N\}}$ in $\{h, t\}$ with outcomes h heads or t tails, whose probability is pinned down by e_U . The last feature is the sequence's share of heads $e_S \in \{\frac{x}{N}\}_{x \in \{0, 1, \dots, N\}}$. Using [equation \(16\)](#), the estimated probability of sequence e is given by:

$$(17) \quad \Pr(e; \alpha_P) \propto \mathbb{P}(e_S | e_N(\alpha_P), \dots, e_U(\alpha_P))^{\alpha_S} \\ \times \prod_{i=1}^N \mathbb{P}(e_i | e_{i-1}(\alpha_P), \dots, e_U(\alpha_P))^{\alpha_i} \mathbb{P}(e_U)^{\alpha_U},$$

which factorizes the distorted probabilities of the selected coin e_U , its realized flips $(e_j)_{j \in \{1, \dots, N\}}$, and the share of heads e_S . Full attention to these features gives the correct estimate.

We consider two categories based on the features we just described, capturing different statistical problems. Two context features play a central role: the coin selection and sequence length, denoted U and N , respectively.

1. Frequency Estimation. This category, *freq*, refers to experiences of estimating the probability of a single draw from a known process, for example, the probability that a fair coin lands on h or t . Among the $N + 2$ features in e , attention α_{freq} focuses on the i.i.d. draw corresponding to the hypothesis, for instance the first flip, $\alpha_1 = 1$ and $\alpha_i = 0$ for all other $i \in \Phi_E$. This category has two diagnostic context features: (i) the coin is fair, $\kappa_{freq, U} = 0.5$ (denoting the probability of h) and (ii) there is a single draw, $\kappa_{freq, N} = 1$.

14. Here the decision rule computes the relative probability of hypotheses as the relative value of two lotteries (as in [Savage 1971](#), section 9, we can allow for incentive-compatible elicitation of multiple events). We can also apply our model to similarity judgments between objects A and B , as in Tversky's famous Austria example ([Tversky 1977](#)); see [Online Appendix B.4](#).

2. *Agnostic Inference.* Category *inf* refers to experiences based on i.i.d. noisy signals without having a DGP prior, for instance, judging the quality of a restaurant based on a few reviews. Attention α_{inf} focuses on the share of positive signals (which in this case is a sufficient statistic for the DGP), setting $\alpha_S = 1$ and $\alpha_i = 0$ for all other $i \in \Phi_E$. The category also has two diagnostic context features: (i) the coin type is not known, $\kappa_{inf,U} = [0, 1]$, and (ii) there are at least two draws (selection of the DGP and one or more signals), $\kappa_{inf,N} \geq 2$.

When solving a statistical problem P , these categories compete for recognition, based on experiences and context. To illustrate, we show how the model generates the gambler's fallacy (GF): people underestimate the probability of obtaining, with a fair coin, an unbalanced sequence such as *hhhhhh* relative to a balanced one of the same length, such as *htthht*. The model clarifies the conditions under which this bias arises, and makes new predictions. (We study biases in inference problems in [Online Appendix B](#).)

3. *Current Problem P.* The atoms of the i.i.d. problem we consider entail a fair coin, $e_U = 0.5$, but specify different flips e_j and share of heads e_S . The hypotheses are two specific atoms (here, the sequences), one balanced and the other not. The context vector κ_P reports that the coin is fair, $\kappa_{P,U} = 0.5$, and that the number of draws is $\kappa_{P,N} = N_P$.¹⁵

The problem does not perfectly fit either category, *freq* or *inf*, that the DM is familiar with. The coin is fair, $\kappa_{P,U} = 0.5$, which matches the frequency category but not inference, but there are many draws, which matches inference but not frequency. We study the problem under the condition that, in the relevant range of N , adding flips makes the distance $d_N(1, N)$ to the single draw characteristic of *freq* grow sufficiently fast; see the [Online Appendix](#) for details.

Based on her categories, the DM can recognize P as *freq* by focusing on the fair coin feature. This recognition draws her attention to the 50:50 nature of flips. Alternatively, the DM can recognize the problem as *inf* by focusing on the length of the sequence. This recognition draws her attention to the share of heads. Denote

15. As in [Section II](#), the context vector can include other features such as “the outcomes of the coin coincide with the hypotheses,” or features of sequences such as “*h* and *t* are alternating.” We abstract from these features because they do not play a role in this application.

by d_N the similarity distance in the length of sequence feature N . With negligible shocks $\epsilon_c, \lambda \rightarrow \infty$, we obtain the following result.

PROPOSITION 5. Suppose that $d_N(N_P + 1, \kappa_{inf, N}) = 0$ for all $N_P \geq$

2. There is a threshold $N\left(\frac{F_{freq}}{F_{inf}}\right)$ with $N(\cdot)$ increasing in its argument, such that the DM recognizes P as *freq* if $N_P < N\left(\frac{F_{freq}}{F_{inf}}\right)$ and as *inf* otherwise. Estimated odds satisfy:

$$(18) \quad \frac{v(htht...htht; \alpha_P^*)}{v(hhhh...hhhh; \alpha_P^*)} = \begin{cases} 1 & \text{if } N_P < N\left(\frac{F_{freq}}{F_{inf}}\right) \\ \frac{\mathbb{P}(e_S=0.5|\Omega)}{\mathbb{P}(e_S=1|\Omega)} > 1 & \text{if } N_P \geq N\left(\frac{F_{freq}}{F_{inf}}\right) \end{cases}.$$

The DM recognizes P as frequency estimation if N_P is low enough. She recognizes P as inference otherwise. Sequence length N_P shapes recognition because it is diagnostic of the two categories. Experiences also matter: higher relative familiarity with *freq*, higher $\frac{F_{freq}}{F_{inf}}$, favors this category over inference.

Critically, if N_P is low, the DM recognizes *freq*, she focuses on the fairness of the coin, and thus she correctly estimates the odds of sequences (despite her misestimation of their absolute probabilities).

In contrast, if N_P is high, then categorization switches to *inf* and the DM focuses on the share of heads, which is indeed relevant in inference. This causes her to represent P as assessing the “share of heads equivalence class” of *htht...htht* and *hhhh...hhhh*. This class is larger for balanced hypotheses, causing the GF. The inference representation causes the GF not because the DM believes the coin type is not known. In fact, she uses the fair coin to compute the probability of e in equation (16). It causes the GF because “inference” prompts the DM to focus on a feature of hypotheses, their share of heads, that causes question substitution. As a consequence, the DM makes an incorrect computation.

Our mechanism for the GF makes new predictions based on sequence length N_P and on other cognitive proxies. The first is experiences F_c : a DM who recently solved many inference problems should be more likely to commit the GF than one who has not, due to lower $\frac{F_{freq}}{F_{inf}}$. The second is salience: making the 50:50 nature of flips more prominent in the choice context should hinder the recognition of the *inf* category and the GF, as shown by Bordalo et al.

(2026a).¹⁶ In [Online Appendix B.2](#), we show that competition between the *freq* and *inf* categories also accounts for multimodality and instability in inference ([Benjamin 2019](#); [Bordalo et al. 2026a](#)).

We shed light on the use of misspecified models neglecting the selection or correlation across sources of information ([Enke and Zimmermann 2019](#); [Enke 2020](#)). This evidence detects heterogeneity (some people account for correlation, others neglect it) and instability (neglect is stronger if sequences are longer, including based on redundant signals). These biases emerge from recognition of inference, which focuses the DM on the share of heads, causing the neglect of the serial correlation. This account is consistent with the bias becoming (i) stronger with longer sequences and (ii) weaker when selection is bottom-up salient. Individual-level heterogeneity may in turn be attributable to experience with different problems.

This logic also sheds light on the use of narratives. People scarred by high inflation ([Malmendier, Nagel, and Yan 2021](#)) may view economic forecasting as “forecasting the costs of inflation,” neglecting contemporaneous low unemployment. The same people can focus on unemployment when it is salient in the media. When thinking about a shock to interest rates, households tend to focus on supply-side mechanisms, which are more familiar to them, but converge to a demand-side view when cued to think about business owners ([Andre et al. 2022](#)). Investors may perceive an electric car company as belonging to the mature automotive sector, but become more optimistic if the firm reframes its business as software, due to shifting attention across the firm’s features ([Sarkar 2025](#)).

V. BOTTOM-UP ATTENTION AND FRAMING

The allocation of attention is not determined exclusively top-down: it is also exogenously shaped by bottom-up salience. People attend more to known taxes when those appear on the price tag ([Chetty, Looney, and Kroft 2009](#)) and prefer goods that are physically present ([Bushong et al. 2010](#)). They also attend more to contrasting payoffs ([Bordalo, Gennaioli, and Shleifer 2012, 2013](#);

16. [Rabin and Vayanos \(2010\)](#) show that a DM who incorrectly believes in the GF may exhibit a nonmonotonic pattern of under-/overreaction depending on whether the DGP is independent or autocorrelated. Their model cannot explain why the same person exhibits the GF on one occasion but not on a normatively equivalent occasion based on changing context.

Kőszegi and Szeidl 2013).¹⁷ The interaction of bottom-up salience and categorization yields a theory of framing: drawing attention to an already known feature of the problem can change how people solve it (Enke and Zimmermann 2019; Bordalo et al. 2026a). These framing effects are readily testable because bottom-up salience can be experimentally controlled.

V.A. Bottom-Up Salience

If a feature is shrouded, it has minimal bottom-up salience. If it is not shrouded, its salience is shaped by sensory prominence and contrast.

Sensory prominence is captured in the description of P , which, like the problem itself, is also an attention and context vector $(\alpha_\delta, \kappa_\delta)$. Subvector κ_δ reports the described context, which captures all that the DM perceives, so $\kappa_P = \kappa_\delta$. It is in part set by nature (e.g., “the sky” is exogenously part of κ_δ in an outdoor task), partly designed (e.g., a monetary lottery could be labeled as a “stock market bet”). Subvector $\alpha_\delta \in [0, 1]^\Phi$ reports the prominence of different features in the description of P . If $\alpha_{\delta,i} = 0$ the feature i is shrouded as in Gabaix and Laibson (2006); if $\alpha_{\delta,i} = 1$ it is “front and center.” Visual prominence can be measured as in Li and Camerer (2022); related methods can be used for text.

Contrast captures attention drawn to highly variable hedonics and events, as in Bordalo, Gennaioli, and Shleifer (2013) and Bordalo et al. (2026a). Goal optimal attention (Simon 1955; Woodford 2012; Gabaix 2019) also depends on payoffs and probabilities, but contrast may excessively focus the DM on striking features, causing the neglect of others. Contrast of $i \in \Phi$ is a real valued function σ_i of that feature’s realizations across atoms, $\sigma_i = \hat{\sigma}_i[(y_i)_{y \in Y}] \in [1, 2]$, with $\hat{\sigma}_i[(y_i)_{y \in Y}] = 1$ if $y_i = y'_i$ for all $y, y' \in Y = \cup_{o \in O}$.¹⁸

Both the problem and the categories are sources of contrast. Contrast from the problem description, denoted $\sigma_{\delta,i}$, compares the values of feature i as described in P , for example, the prices of different jams in the current supermarket. Contrast from category c , denoted $\sigma_{c,i}$, compares the values of i in experiences in c , such as variation in past jam prices. Often these measures overlap, but a

17. See Lanzani (2022) for the revealed-preference axioms underpinning these effects.

18. This nests the contrast function in Bordalo, Gennaioli, and Shleifer (2013); see Online Appendix B for details.

feature can have a high category contrast—and be attended to—even if it is currently shrouded: before flying, some people think about crashes because they represent a contrasting payoff from memory. Contrast is computed using context vectors κ_δ and κ_c , which report the values of hedonic and event features.¹⁹ Context features themselves have minimal contrast, because they do not vary across options.

In [Section II](#), the DM sets attention to maximize the similarity of the representation (α_P, κ_P) to the category (α_c, κ_c) . Here, the DM allocates attention across features to trade off the problem's similarity to the category versus the description $(\alpha_\delta, \kappa_P)$. [Equation \(6\)](#) then becomes:

$$(20) \quad (\alpha_P^*, c^*) = \operatorname{argmax}_{\alpha_P \in [0,1]^\Phi, c \in C} F_c \cdot S[(\alpha_P, \kappa_P), (\alpha_c, \kappa_c) | \sigma_c] \\ + S[(\alpha_P, \kappa_P), (\alpha_\delta, \kappa_P) | \sigma_\delta] + \epsilon_c,$$

where the category-similarity function attaches higher weights to hedonics and events with more category contrast, and similarly for description-similarity ([Reed 1972](#); [Nosofsky 1986](#)):

$$(21) \quad S[(\alpha_P, \kappa_P), (\alpha_x, \kappa_x) | \sigma_x] = \\ 1 - \frac{\sum_{i \in \Phi} \sigma_{x,i} d(|\alpha_{P,i} - \alpha_{x,i}|) + \sum_{i \in \Phi_K} \alpha_{P,i} \alpha_{x,i} d_i(\kappa_{P,i}, \kappa_{x,i})}{|\Phi| + |\Phi_K|} \\ \forall x \in \{\delta, c\}.$$

We show the role of bottom-up salience using our consumption example (see also [Proposition B.1](#) in [Online Appendix B.1](#)). We then shed light on well-known framing effects in lottery choice.

V.B. Revisiting the Context-Dependent Consumer

The riskless consumer problem studied in [Section III](#), based on neglect of the bad state b , raises two questions. First, what determines neglect versus attention to the low-probability risk b ? Unlikely events are often ignored, as in our example, but at other times they are overweighted, as in [Kahneman and Tversky \(1979\)](#). Second, context is not the only driver of instability. Advertising and negotiation change choices by emphasizing some relevant features at the expense of others. How do such interventions work?

19. If the feature is shrouded, $\alpha_{\delta,i} = 0$, its values are not encoded in κ_δ . In this case, description contrast is minimal, $\sigma_{\delta,i} = 1$. Similarly, category contrast is minimal, $\sigma_{c,i} = 1$, if the feature was neglected in c , $\alpha_{c,i} = 0$.

These questions are linked to bottom-up salience. Making a feature more bottom-up salient frames the problem in two ways: (i) it affects attention to that feature in each category, and (ii) it shapes categorization. We first explore how a frame shapes sensitivity to attributes in a category.

Let $\alpha_{P,i}(c|\delta)$ be the attention to feature i in a given category c when prominence is α_δ . This attention is found by maximizing similarity in [equation \(20\)](#) at fixed c .

PROPOSITION 6. Attention $\alpha_{P,i}(c|\delta)$ to feature $i \in \Phi_H \cup \Phi_E$ increases in its prominence $\alpha_{\delta,i}$ and in category c 's attention $\alpha_{c,i}$.²⁰

The description is more influential, that is, $\inf_{\alpha_i \in \alpha_{P,i}(c|\delta)} |\alpha_i - \alpha_{\delta,i}|$ is lower, when $\frac{\sigma_{c,i}F_c}{\sigma_{\delta,i}}$ is lower. In particular, if $d(x) = x$ for all $x \in \mathbb{R}_+$, we have:

$$(22) \quad \alpha_{P,i}(c|\delta) = \begin{cases} \alpha_{\delta,i} & \text{if } \frac{\sigma_{c,i}F_c}{\sigma_{\delta,i}} < 1 \\ \alpha_{c,i} & \text{if } \frac{\sigma_{c,i}F_c}{\sigma_{\delta,i}} > 1 \end{cases} \quad \forall i \in \Phi_H \cup \Phi_E.$$

Increasing a feature's bottom-up salience weakly increases the attention it receives independently of recognition. By [equation \(22\)](#), this effect is strongest if $\frac{\sigma_{c,i}F_c}{\sigma_{\delta,i}} < 1$: feature i is attended to when it is prominent ($\alpha_{\delta,i} = 1$) and neglected when it is shrouded ($\alpha_{\delta,i} = 0$), even if the category prescribes otherwise. This occurs if the DM is either unfamiliar with c (F_c is low), or if the category contrast of feature i is small compared with its bottom-up contrast ($\frac{\sigma_{c,i}}{\sigma_{\delta,i}}$ is low). Indeed, making quality prominent, $\alpha_{\delta,Qg} = 1$, promotes attention to it, especially if qualities on offer are contrasting, high $\sigma_{\delta,Qg}$, as in [Bordalo, Gennaioli, and Shleifer \(2012, 2013\)](#).

Conversely, bottom-up salience does not matter if the category is a strong driver of attention to a feature, $\frac{\sigma_{c,i}F_c}{\sigma_{\delta,i}} > 1$. This can occur because the category is highly familiar, high F_c , or because the feature had high contrast in the past, high $\frac{\sigma_{c,i}}{\sigma_{\delta,i}}$. A person with many poverty experiences will attend to price even if it is shrouded, due to high F_{buy} and to past opportunity costs, high $\sigma_{c,M}$.

Bottom-up salience sheds light on our first question, namely, why are rare extreme payoffs sometimes neglected? This occurs when such payoffs are shrouded and have not been experienced in the past, in which case bottom-up salience and categorization jointly promote a riskless *con* or *buy* representation.

20. The increase in $\alpha_{P,i}(c|\delta)$ is with respect to the strong set order.

Changes in bottom-up salience then shed light on our second question: why do representations sometimes change with irrelevant advertising and salience? This occurs when bottom-up attention is drawn to previously neglected payoffs, particularly if they are extreme, high $\sigma_{\delta, Qb}$. Crucially, this effect can cause a shift in categorization, giving a theory of framing effects. We illustrate this logic in our running example, and study the general case in the [Online Appendix](#).²¹

Suppose that a batch of jam has been recalled because of contamination concerns. The media extensively describe the health risks, boosting the prominence $\alpha_{\delta, Qb}$ of the bad outcome q_b , which is severe compared with normal quality, $q_g + q_b < 0$. The problem can be represented as *consuming* a jam, as *buying* it, or as “estimating the probability of a bad batch”, using the *freq* category. To see which representations prevails, assume that the context κ_{freq} associated to *freq* in the past is distant from current context κ_P , so the consumer still chooses between *con* and *buy*, as in [Section III](#). [Proposition 6](#) then implies the following result.

PROPOSITION 7. Suppose that context is sufficiently close to buying (κ_P low) that the consumer originally adopts a riskless *buy* category. As the prominence $\alpha_{\delta, Qb}$ of Qb increases from zero to one:

- i. Similarity of P to *buy* falls while that to *con* increases.
- ii. Suppose that $d(x) = x$ for all $x \in [0, 1]$, $\alpha_{\delta, W} = 0$, $\frac{\sigma_{con, W} F_{con}}{\sigma_{\delta, W}} < 1$, and that $\frac{\sigma_{con, i} F_{con}}{\sigma_{\delta, i}} > 1$ for all $i \in \Phi_H$. If the consumer switches to *con*, her willingness to pay falls to $\frac{q_g + q_b}{2\alpha\eta}$.

As in [Proposition 6](#), making contamination concerns salient draws attention to the bad payoff in any category. Critically, and this is where reframing occurs, this description is more consistent with *con*, where qualities (good or bad) are fully attended to, than with *buy*, where quality shocks are neglected. The prominence of risk promotes recategorization into *con*, leading to instability: not

21. Proposition B.1 in the [Online Appendix](#) shows this result holds in general. These forces help explain the experience-description gap in lottery choice ([Hertwig and Erev 2009](#)). They also produce the tendency to project today's tastes into the future ([Loewenstein, O'Donoghue, and Rabin 2003](#)). [Bordalo, Gennaioli, and Shleifer \(2020\)](#) present a memory model in which the current experience helps simulate a similar future. Our framework suggests a related channel: the current experience is prominent so it receives a high weight in valuation.

only does the consumer now focus on the downside q_b , she also reduces her focus on price and neglects the fact that the probability $\mathbb{P}(b)$ is low. As a consequence, she exhibits (i) pessimistic beliefs about quality, and (ii) a lower price sensitivity. Together, these imply that firms cannot prop up demand by cutting prices, even by a lot.

A first key implication of [Proposition 7](#) is to link information provision and framing. In a rational model, information about q_b causes a small drop in valuation, due to its small objective probability $\mathbb{P}(b)$. In prospect theory, a low $\mathbb{P}(b)$ is exaggerated, but price sensitivity does not vary with information.²² In our model, in contrast, information causes instability of attention across a range of features, including the downside risk, its probability, and monetary incentives. Information provision affects decisions more by changing context than its news content.

The well-known “certainty effect” ([Kahneman and Tversky 1979](#)) is an instance of this mechanism: adding a rare but high-contrast downside risk discontinuously reduces the valuation of a sure thing ([Haigh and List 2005](#); [Barseghyan et al. 2013](#)) because it directs attention toward extreme payoffs, not toward small probabilities.²³

A second implication of [Proposition 7](#) is that the problem’s description α_δ matters if it raises the prominence of a feature that is “important,” in the sense that it is either attended to in a frequent category or that it varies sharply across options. Framing effects are not random psychology or residual noise. They rely on problem recognition and as such are the strongest when they change the salience of important features.

22. [Proposition 7](#) implies that recategorization requires a sufficiently large increase in the salience of risks. People often shrug when told about risks, as long as they are not made prominent enough.

23. When choosing between \$100 and \$99 categorization into *con* correctly leads to the choice of \$100. Adding a small risk ε of a \$0 payoff to the \$100 payoff introduces a downside feature with contrasting values \$0 and \$99 across options. As it draws attention, it reinforces categorization into *con*, prompting neglect of probabilities and risk aversion. [Bordalo, Gennaioli, and Shleifer \(2013\)](#) study the role of payoff contrast but their continuous payoff weights foreclose discontinuities. Another important example of discontinuity arises in situations involving social norms. [Gneezy and Rustichini \(2000\)](#) show that a small payment reduces effort in the collection of donations, presumably because the payment is now categorized as a low-salary job.

V.C. *Instability in Lottery Choice*

A large body of work on lottery choice in the lab demonstrates systematic instability of risk attitudes relative to the expected utility benchmark, typically explained using prospect theory's reference points or probability weighting functions (Kahneman and Tversky 1979). Our model accounts for these anomalies, revealing deep connections between different choice domains—probability judgments and choice among goods—and yielding new predictions. We illustrate these ideas with several examples, leaving a full treatment to future work.

Lottery choices in lab experiments entail prominent payoffs and probabilities. The DM is torn between a “payoff evaluation” or a “frequency estimation” problem. The first representation is tied to the *con* category and attention to payoff trade-offs. The second is tied to the *freq* category and attention to probability trade-offs. Experiences in *con* are arguably more frequent, but they occur in settings that are less similar to the abstract context of the lab. Instability in categorization, due to variations in bottom-up salience, entails instability of risk attitudes.

1. *Payoff Prominence.* Risk attitudes change with lottery descriptions even if payoffs and probabilities are held constant. Suppose a DM chooses between a lottery paying $m_g > 0$ with probability $\mathbb{P}(g)$, and $m_b = 0$ otherwise, and a sure thing paying its expected value $m = m_g \cdot \mathbb{P}(g)$, which delivers the same payoff m in states g and b . Consider two descriptions of the risky lottery:

(23) Shrouded Downside : win m_g with probability $\mathbb{P}(g)$.

Full Prominence : win m_g with probability $\mathbb{P}(g)$

(24) and 0 otherwise.

In equation (23), the downside payoff and its probability are implied but not prominently described, $\alpha_{\delta, M_b} < 1$ and $\alpha_{\delta, W} < 1$. In equation (24), the prominence of the downside payoff increases to $\alpha_{\delta, M_b} = 1$. This has two effects. First, it increases attention to the downside in any category (with a logic similar to the one of Proposition 6). Second, it promotes adoption of *con* (Proposition 7). As a result, in equation (4) the DM attaches a higher weight α_{P, M_b} to the downside payoff for both the lottery and the sure thing, which favors the safe choice. A mere increase in descriptive promi-

nence of the downside risk increases risk aversion even though the lottery has not changed.²⁴

This effect of prominence is absent from existing theories of choice under risk but can be important in the real world, where prominence varies across payoffs and over time. To take a few examples, advertising high financial returns promotes investment and the neglect of risk, especially when the crash event is shrouded (Mullainathan, Schwartzstein, and Shleifer 2008; Célérier and Vallée 2017).²⁵ Advertising top prizes drives the demand for lottery tickets (Kearney 2005). Demand for insurance coverage rises after uninformative but superficially similar shocks (Kunreuther 1978; Dessaint and Matray 2017). Different parties to a transaction or negotiation emphasize gains or losses differently (Kahneman and Lovallo 1993).

2. Contrast of Payoffs and Probabilities. We illustrate how changing payoffs or probabilities can change their contrast, shift attention across features, and cause recategorization. Consider the Allais common-ratio effect (Allais 1953; Kahneman and Tversky 1979). If people are indifferent between \$100 with probability 0.2 and \$20 with probability 0.8, then they likely prefer \$100 with probability 0.02 to \$20 with probability 0.08. This pattern violates expected utility, where preferences should be invariant to uniform scaling of probabilities.

Our model applies as follows. There is a good lottery state g and a bad state b . The lotteries deliver different payoffs in g , creating payoff contrast σ_{δ, M_g} . They also differ in the probability of state g , creating event contrast $\sigma_{\delta, e}$. Crucially, event contrast changes with the scaling of probabilities: $\sigma_{\delta, e}$ is high when comparing probabilities 0.2 and 0.8 but declines dramatically when comparing 0.02 to 0.08. Proposition B.1 in [Online Appendix B.1](#)

24. Formally, valuation of lottery l and of sure thing s in category c are, respectively:

$$\begin{aligned}\tilde{v}(l, c) &= \mathbb{P}(g)^{\alpha_{PW}(c)} \cdot \alpha_{P, M_g}(c) \cdot m_g, \\ \tilde{v}(s, c) &= \left[\mathbb{P}(g)^{\alpha_{PW}(c)} \cdot \alpha_{P, M_g}(c) + (1 - \mathbb{P}(g))^{\alpha_{PW}(c)} \cdot \alpha_{P, M_b}(c) \right] \cdot m.\end{aligned}$$

Higher attention to the downside payoff $\alpha_{P, M_b}(c)$ boosts the valuation of the sure thing, not of the lottery, promoting the safe choice.

25. A related treatment varies the description of the sure thing by explicitly mentioning that it never yields a downside (unlike the risky alternative), or by keeping it implicit. We thank Alex Imas for making this point as a discussant.

shows that this decline in $\sigma_{\delta,e}$ reduces attention to probabilities and promotes a shift toward the “payoff estimation” representation *con*, which also increases attention to payoffs. This change in weights benefits the riskier lottery, generating the common-ratio Allais paradox.²⁶

When differences in probabilities are large, the DM attends to them and neglects payoffs, encouraging risk aversion. When differences in probabilities are “peanuts,” the DM neglects them and attends to payoffs, encouraging risk taking. This shift in representations offers a foundation for [Rubinstein’s \(1988\)](#) intuition that choice depends on perceived similarity between probabilities.

Instability in risk attitudes driven by probability contrast illuminates risky choices in the field. For instance, in baseball and basketball, players routinely decide whether to favor riskier, high-payoff shots (home runs and three-pointers, respectively). But while home runs were overvalued in baseball prior to *Moneyball* ([Hakes and Sauer 2006](#)), basketball players have been reluctant to take three-point shots despite their higher expected value ([Schwartzstein and Malhotra 2021](#)).

The Allais common-ratio logic can shed light on this behavior. In basketball, the safer two-pointer has a hit rate of 50% compared with 35% for a three-pointer. In baseball, the safer “walk” has a hit rate of 9% compared with 3.5% for the home run. While a basketball two-pointer is $\frac{50}{35} = 1.43$ times more likely to succeed than a three-pointer, in baseball a walk is $\frac{9}{3.5} = 2.57$ times more likely to succeed than a home run. At given payoffs, this logic favors more risk taking in basketball than in baseball, contrary to historical practice.

Our model explains the observed historical practice because the difference in hit rates is much larger in basketball than in baseball (15 versus 5.5 percentage points). Larger probability differences draw attention in basketball, favoring the safer two-

26. Denote the safer lottery by $(\mathbb{P}(w), w)$ and the mean-preserving spread by $(\mathbb{P}(x), x)$. Then, the valuation gap $\tilde{v}(c)$ between the lotteries in category c is given by (see [Online Appendix B.3](#)):

$$\tilde{v}(c) = \mathbb{P}(x)^{\alpha_{P,e}(c)} \cdot \alpha_{P,u}(c) \cdot (x - w) + \left[\mathbb{P}(x)^{\alpha_{P,e}(c)} - \mathbb{P}(w)^{\alpha_{P,e}(c)} \right] \cdot w \cdot \alpha_{P,u}(c).$$

Since $\mathbb{P}(x) \cdot x = \mathbb{P}(w) \cdot w$, the right side increases in attention to payoffs $\alpha_{P,u}(c)$ and decreases in attention to probabilities $\alpha_{P,e}(c)$. In particular, categorization in *con* boosts risk taking while categorization in *freq* boosts risk aversion.

pointer, while small probability differences are neglected in baseball, favoring the home run.²⁷

Last, in [Online Appendix B.3](#), we show that categorization into *con* due to high-contrast payoff differences also explains the fourfold pattern in lotteries, as well as in riskless mirrors ([Oprea 2024; de Clippel, Oprea, and Rozen 2025](#)).²⁸ A common cognitive mechanism of attention driven by categorization and bottom-up salience unifies biases in riskless choice, probability judgments, and risky choice.

VI. CONCLUSION

Canonical models of decision-making, both neoclassical and behavioral, hinge on the maximization of a stable objective function that combines hedonics and probabilities. We present a new approach in which the DM first draws on selective attention and memory to mentally represent the problem at hand, then combines the hedonics and probabilities that are relevant in that representation. As a consequence, the objective function that the DM maximizes exhibits spurious heterogeneity and instability based on past experiences, context, and framing. In contrast to neoclassical and standard behavioral models, this approach unifies observed heterogeneity and instability in judgments and decision-making.

Our analysis raises two important methodological issues. First, to properly understand behavior, cognitive forces such as attention and memory must be formalized and measured. Choice analysis, even if enriched with psychological carriers of utility, is insufficient. Our model makes precise predictions based on cognitive inputs such as context, experiences, and similarity (see [Malmendier and Nagel 2016; Bordalo et al. 2025a, 2026a](#)). It

27. Payoffs move in the same direction of making the risky strategy more appealing in baseball than basketball. One home run is worth 4 to 16 walks, which is much larger than the difference between two and three points in basketball.

28. For similar reasons, our model helps explain weak within-person correlation of choices in a domain (e.g., low correlation between insurance demand and lottery choice; [Barseghyan, Prince, and Teitelbaum 2011](#)) and across domains (e.g., low correlation between the endowment effect and aversion to mixed lotteries; [Chapman et al. 2024](#)). Although in this article we limit ourselves to the case of known probabilities, the same mechanism could connect the striking relation between behavior facing ambiguity and decisions with known probabilities documented in [Halevy \(2007\)](#).

also opens the door for a structured use of AI methods, which can recover representations from self-reported reasons for choice (Haaland et al. 2024; Link et al. 2025), as well as unveil subtle contextual features, as in Ludwig and Mullainathan (2024). Future work can use such data in a theoretically disciplined way to study the cognitive structure of decisions.

The second methodological point concerns experimental design. Current practice favors the use of abstract protocols to identify “universal” choice biases and minimize experimenter demand. Abstraction can help capture general cognitive forces but is problematic for studying real-world choices, such as demand for insurance, for two reasons. First, removing naturalistic context can also remove consumers’ spontaneous representations, reducing external validity for predicting field behavior. Second, elaborate and abstract protocols geared at measuring “true” tastes may anchor assessments to laboratory context features, providing a distorted metric of normative decision quality. Naturalistic immersion is a benefit of field experiments, but our theory suggests that future work should also develop methods to properly engineer controlled variation of naturalistic context in the lab.

Our analysis opens several new avenues, and we conclude with two of them. The first concerns the nature and origin of categories. We showed that two riskless choice and two statistical categories suffice to unify a great deal of evidence. They could also help explain evidence we did not discuss. For instance, by focusing people on the cost of giving away resources, the *buy* category may help account for phenomena related to loss aversion, such as reluctance to take the risk of losing money or the unwillingness to pay to acquire a good.²⁹ Of course, there are many more categories. In risky choice, a natural category can entail regret or elation in which attention is focused on these features of experience. In intertemporal choice, a natural category is one of “investment” decisions, in which attention is focused on long-term payoffs. Future work should study where such categories come from. One possibility is that they are formed through bottom-up salient features of experiences, such as a feeling, a striking payoff, or being exposed to social reminders of cultural norms, which then become diagnostic markers for different categories of problems. Because such bottom-up

29. In this way, we could endogenize the utility function in Rabin and Weizsäcker (2009), which combines integrated and nonintegrated lotteries, as a function of experience and description.

forces are potentially measurable, endogenous categories would enhance the explanatory power of our approach.

A second major avenue is to study the real-world implications of our mechanism. When thinking about redistribution, some voters may think about fairness, others about zero-sum transfers from taxpayers (Chinoy et al. 2026) based on different experiences, but changes in context such as the specific name or domain of the tax may change problem similarity, representations, and voter preferences. Some people think guns at home bring safety, others that they bring the risk of an accident (Alsan, Schwartzstein, and Stantcheva 2026). Similar considerations apply to fairness judgments (Kahneman, Knetsch, and Thaler 1986), which may be viewed from the perspective of equity or of endowment, to strategic behavior (List 2007), which may be viewed in terms of maximizing self-interest or punishing transgressive behavior, and many other settings.

Ongoing work explores the role of reasoning in real-world settings that also require estimation of payoffs and probabilities. Bordalo et al. (2026a) show that priming individuals to recall personal adversity shifts their thinking about inflation away from economic news and toward concerns about affordability and personal distress, leading them to expect higher inflation. Such real-world applications are a key testing ground not only of the model's explanatory power but also of the design of choice architectures to improve mental representations and choices.

SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

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REFERENCES

- Allais, Maurice, "Le Comportement de l'Homme Rationnel devant le Risque: Critique des Postulats et Axiomes de l'Ecole Americaine," *Econometrica*, 21 (1953), 503–546. <https://doi.org/10.2307/1907921>.
- Alsan, Marcella, Joshua Schwartzstein, and Stefanie Stantcheva, "The Universal Pursuit of Safety and the Demand for (Lethal, Non-Lethal or No) Guns," Work-

- ing Paper no. 34962, National Bureau of Economic Research, Cambridge, MA, 2026. <https://doi.org/10.3386/w34962>.
- Andre, Peter, Carlo Pizzinelli, Christopher Roth, and Johannes Wohlfart, "Subjective Models of the Macroeconomy: Evidence from Experts and Representative Samples," *Review of Economic Studies*, 89 (2022), 2958–2991. <https://doi.org/10.1093/restud/rdac008>.
- Andre, Peter, Christopher Roth, Ingar Haaland, and Johannes Wohlfart, "Inflation Narratives," CEPR Discussion Paper No. DP16758, 2021. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4026601.
- Awh, Edward, Artem V. Belopolsky, and Jan Theeuwes, "Top-Down Versus Bottom-Up Attentional Control: A Failed Theoretical Dichotomy," *Trends in Cognitive Sciences*, 16 (2012), 437–443. <https://doi.org/10.1016/j.tics.2012.06.010>.
- Ba, Cuimin, J. Aislinn Bohren, and Alex Imas, "Over- and Underreaction to Information," available at SSRN 4274617, 2024. <https://dx.doi.org/10.2139/ssrn.4274617>.
- Baicker, Katherine, Sendhil Mullainathan, and Joshua Schwartzstein, "Behavioral Hazard in Health Insurance," *Quarterly Journal of Economics*, 130 (2015), 1623–1667. <https://doi.org/10.1093/qje/qjv029>.
- Banerjee, Abhijit V., and Esther Duflo, "The Economic Lives of the Poor," *Journal of Economic Perspectives*, 21 (2007), 141–168. <https://doi.org/10.1257/jep.21.1.141>.
- Barseghyan, Levon, Francesca Molinari, Ted O'Donoghue, and Joshua C. Teitelbaum, "The Nature of Risk Preferences: Evidence from Insurance Choices," *American Economic Review*, 103 (2013), 2499–2529. <https://doi.org/10.1257/aer.103.6.2499>.
- Barseghyan, Levon, Jeffrey Prince, and Joshua C. Teitelbaum, "Are Risk Preferences Stable across Contexts? Evidence from Insurance Data," *American Economic Review*, 101 (2011), 591–631. <https://doi.org/10.1257/aer.101.2.591>.
- Bassok, Miriam, "Analogical Transfer in Problem Solving," in *The Psychology of Problem Solving*, J. E. Davidson and R. J. Sternberg, ed. (Cambridge: Cambridge University Press, 2003), 343–369. <https://doi.org/10.1017/CBO9780511615771.012>.
- Bastianello, Francesca, Paul Décaire, and Marius Guenzel, "Mental Models and Financial Forecasts," Jacobs Levy Equity Management Center for Quantitative Financial Research Paper, 2024. <https://doi.org/10.2139/ssrn.5004839>.
- Becker, Gary S., and Kevin M. Murphy, "A Simple Theory of Advertising as a Good or Bad," *Quarterly Journal of Economics*, 108 (1993), 941–964. <https://doi.org/10.2307/2118455>.
- Benjamin, Daniel J., "Errors in Probabilistic Reasoning and Judgment Biases," in *Handbook of Behavioral Economics: Applications and Foundations 1*, vol. 2, B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, eds. (Amsterdam: North-Holland, 2019), 69–186. <https://doi.org/10.1016/bs.hesbe.2018.11.002>.
- Bordalo, Pedro, Giovanni Burro, Katherine Coffman, Nicola Gennaioli, and Andrei Shleifer, "Imagining the Future: Memory, Simulation and Beliefs," *Review of Economic Studies*, 92 (2025a), 1532–1563. <https://doi.org/10.1093/restud/rdae070>.
- Bordalo, Pedro, Giovanni Burro, Nicola Gennaioli, Gad Nacamura, and Andrei Shleifer, "Ads as Cues," Working Paper no. 34387, National Bureau of Economic Research, Cambridge, MA, 2025b. <https://doi.org/10.3386/w34387>.
- Bordalo, Pedro, John J. Conlon, Nicola Gennaioli, Spencer Y. Kwon, and Andrei Shleifer, "Memory and Probability," *Quarterly Journal of Economics*, 138 (2023), 265–311. <https://doi.org/10.1093/qje/qjac031>.
- Bordalo, Pedro, John Conlon, Nicola Gennaioli, Spencer Kwon, and Andrei Shleifer, "How People Use Statistics," *Review of Economic Studies*, 93 (2026a), 250–285. <https://doi.org/10.1093/restud/rdaf022>.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer, "Competition for Attention," *Review of Economic Studies*, 83 (2016), 481–513. <https://doi.org/10.1093/restud/rdv048>.

- , “Memory, Attention, and Choice,” *Quarterly Journal of Economics*, 135 (2020), 1399–1442. <https://doi.org/10.1093/qje/qjaa007>.
- , “Salience and Consumer Choice,” *Journal of Political Economy*, 121 (2013), 803–843. <https://doi.org/10.1086/673885>.
- , “Salience Theory of Choice Under Risk,” *Quarterly Journal of Economics*, 127 (2012), 1243–1285. <https://doi.org/10.1093/qje/qjs018>.
- , “Salience,” *Annual Review of Economics*, 14 (2022), 521–544. <https://doi.org/10.1146/annurev-economics-051520-011616>.
- Bordalo, Pedro, Nicola Gennaioli, Rafael La Porta, and Andrei Shleifer, “Belief Overreaction and Stock Market Puzzles,” *Journal of Political Economy*, 132 (2024), 1450–1484. <https://doi.org/10.1086/727713>.
- , “Finance Without Exotic Risk,” *Journal of Financial Economics*, 173 (2025c), 104145. <https://doi.org/10.1016/j.jfineco.2025.104145>.
- Bordalo, Pedro, Nicola Gennaioli, Florencio Lopez de Silanes, Simon G. Schröder, Andrei Shleifer, and Maarten van Rooij, “The Psychology of Macroeconomic Expectations,” Working Paper No. 35214, National Bureau of Economic Research, Cambridge, MA, 2026b. <https://doi.org/10.3386/w35214>.
- Bushong, Benjamin, Lindsay M. King, Colin F. Camerer, and Antonio Rangel, “Pavlovian Processes in Consumer Choice: The Physical Presence of a Good Increases Willingness-to-Pay,” *American Economic Review*, 100 (2010), 1556–1571. <https://doi.org/10.1257/aer.100.4.1556>.
- Camerer, Colin, and Eric Johnson, “The Process Performance Paradox in Expert Judgment,” in *Toward a General Theory of Expertise*, A. Ericsson and J. Smith, eds. (Cambridge: Cambridge University Press, 1991).
- Célérier, Claire, and Boris Vallée, “Catering to Investors Through Security Design: Headline Rate and Complexity,” *Quarterly Journal of Economics*, 132 (2017), 1469–1508. <https://doi.org/10.1093/qje/qjx007>.
- Cenzon, Josefina, “Credit Market Experiences and Macroeconomic Expectations: Evidence and Theory,” Working Paper, Bocconi University, 2025.
- Chandra, Amitabh, Evan Flack, and Ziad Obermeyer, “The Health Costs of Cost Sharing,” *Quarterly Journal of Economics*, 139 (2024), 2037–2082. <https://doi.org/10.1093/qje/qjae015>.
- Chapman, Jonathan, Mark Dean, Pietro Ortoleva, Erik Snowberg, and Colin Camerer, “Willingness to Accept, Willingness to Pay, and Loss Aversion,” Working Paper no. 30836, National Bureau of Economic Research, Cambridge, MA, 2024. <https://doi.org/10.3386/w30836>.
- Chase, William G., and Herbert A. Simon, “Perception in Chess,” *Cognitive Psychology*, 4 (1973), 55–81. [https://doi.org/10.1016/0010-0285\(73\)90004-2](https://doi.org/10.1016/0010-0285(73)90004-2).
- Chetty, Raj, Adam Looney, and Kory Kroft, “Salience and Taxation: Theory and Evidence,” *American Economic Review*, 99 (2009), 1145–1177. <https://doi.org/10.1257/aer.99.4.1145>.
- Chinoy, Sahil, Nathan Nunn, Sandra Sequeira, and Stefanie Stantcheva, “Zero-Sum Thinking and the Roots of US Political Divides,” *American Economic Review*, 116 (2026), 1052–1096. <https://doi.org/10.1257/aer.20240692>.
- de Clippel, Geoffroy, Ryan Oprea, and Kareen Rozen, “As If,” Working Paper, Brown University, 2025.
- Dessaint, Olivier, and Adrien Matray, “Do Managers Overreact to Salient Risks? Evidence from Hurricane Strikes,” *Journal of Financial Economics*, 126 (2017), 97–121. <https://doi.org/10.1016/j.jfineco.2017.07.002>.
- Edwards, Ward, “Conservatism in Human Information Processing,” in *Judgment under Uncertainty: Heuristics and Biases*, Kahneman Daniel, Slovic Paul, and Tversky Amos, eds. (Cambridge: Cambridge University Press, 1968), 359–369.
- Ellis, Andrew, and Yusufcan Masatlioglu, “Choice with Endogenous Categorization,” *Review of Economic Studies*, 89 (2022), 240–278. <https://doi.org/10.1093/restud/rdab011>.
- Enke, Benjamin, “What You See Is All There Is,” *Quarterly Journal of Economics*, 135 (2020), 1363–1398. <https://doi.org/10.1093/qje/qjaa012>.

- Enke, Benjamin, and Thomas Graeber, "Cognitive Uncertainty," *Quarterly Journal of Economics*, 138 (2023), 2021–2067. <https://doi.org/10.1093/qje/qjad025>.
- Enke, Benjamin, and Florian Zimmermann, "Correlation Neglect in Belief Formation," *Review of Economic Studies*, 86 (2019), 313–332. <https://doi.org/10.1093/restud/rdx081>.
- Fan, Tingzhe, Yueran Liang, and Cameron Peng, "The Inference-Forecast Gap in Belief Updating," *Econometrica*, forthcoming.
- Frederick, Shane, Nathan Novemsky, Jing Wang, Ravi Dhar, and Stephen Nowlis, "Opportunity Cost Neglect," *Journal of Consumer Research*, 36 (2009), 553–561. <https://doi.org/10.1086/599764>.
- Fryer, Roland, and Matthew Jackson, "A Categorical Model of Cognition and Biased Decision Making," *BE Journal of Theoretical Economics*, 8 (2008).
- Fudenberg, Drew, Giacomo Lanzani, and Philipp Strack, "Selective-Memory Equilibrium," *Journal of Political Economy*, 132 (2024), 3978–4020. <https://doi.org/10.1086/731412>.
- , "Learning, Memory, and Stochastic Choice," available at SSRN 4923469, 2025.
- Gabaix, Xavier, "Behavioral Inattention," in *Handbook of Behavioral Economics: Applications and Foundations 1*, vol. 2, B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, eds. (Amsterdam: North-Holland, 2019), 261–343. <https://doi.org/10.1016/bs.hesbe.2018.11.001>.
- Gabaix, Xavier, and David Laibson, "Shrouded Attributes, Consumer Myopia, and Information Suppression in Competitive Markets," *Quarterly Journal of Economics*, 121 (2006), 505–540. <https://doi.org/10.1162/qjec.2006.121.2.505>.
- Gagnon-Bartsch, Tristan, Matthew Rabin, and Joshua Schwartzstein, "Channeled Attention and Stable Errors," Mimeo, Harvard University, 2023.
- Gennaioli, Nicola, Marta Leva, Raphael Schoenle, and Andrei Shleifer, "How Inflation Expectations De-anchor: The Role of Selective Memory Cues," Working Paper no. 32633, National Bureau of Economic Research, Cambridge, MA, 2024. <https://doi.org/10.3386/w32633>.
- Gilboa, Itzhak, and David Schmeidler, "Case-Based Decision Theory," *Quarterly Journal of Economics*, 110 (1995), 605–639. <https://doi.org/10.2307/2946694>.
- Gneezy, Uri, and Aldo Rustichini, "Pay Enough or Don't Pay at All," *Quarterly Journal of Economics*, 115 (2000), 791–810. <https://doi.org/10.1162/003355300554917>.
- Goldstone, Robert L., "Influences of Categorization on Perceptual Discrimination," *Journal of Experimental Psychology: General* 123 (1994), 178–200. <https://doi.org/10.1037/0096-3445.123.2.178>.
- Gourville, John T., and Dilip Soman, "Payment Depreciation: The Behavioral Effects of Temporally Separating Payments from Consumption," *Journal of Consumer Research*, 25 (1998), 160–174. <https://doi.org/10.1086/209533>.
- Green, Etan A., Justin M. Rao, and David Rothschild, "A Sharp Test of the Portability of Expertise," *Management Science*, 65 (2019), 2820–2831. <https://doi.org/10.1287/mnsc.2018.3063>.
- Grether, David M., "Bayes Rule as a Descriptive Model: The Representativeness Heuristic," *Quarterly Journal of Economics*, 95 (1980), 537–557. <https://doi.org/10.2307/1885092>.
- Haaland, Ingar K., Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart, "Measuring What Is Top of Mind," Working Paper no. 32421, National Bureau of Economic Research, Cambridge, MA, 2024. <https://doi.org/10.3386/w32421>.
- Haigh, Michael S., and John A. List, "Do Professional Traders Exhibit Myopic Loss Aversion? An Experimental Analysis," *Journal of Finance*, 60 (2005), 523–534. <https://doi.org/10.1111/j.1540-6261.2005.00737.x>.
- Hakes, Jahn K., and Raymond D. Sauer, "An Economic Evaluation of the Moneyball Hypothesis," *Journal of Economic Perspectives*, 20 (2006), 173–186. <https://doi.org/10.1257/jep.20.3.173>.

- Halevy, Yoram, "Ellsberg Revisited: An Experimental Study," *Econometrica*, 75 (2007), 503–536. <https://doi.org/10.1111/j.1468-0262.2006.00755.x>.
- Henderson, Pamela W., and Robert A. Peterson, "Mental Accounting and Categorization," *Organizational Behavior and Human Decision Processes*, 51 (1992), 92–117. [https://doi.org/10.1016/0749-5978\(92\)90006-S](https://doi.org/10.1016/0749-5978(92)90006-S).
- Hertwig, Ralph, and Ido Erev, "The Description-Experience Gap in Risky Choice," *Trends in Cognitive Sciences*, 13 (2009), 517–523. <https://doi.org/10.1016/j.tics.2009.09.004>.
- Hoch, Stephen J., Byung-Do Kim, Alan L. Montgomery, and Peter E. Rossi, "Determinants of Store-Level Price Elasticity," *Journal of Marketing Research*, 32 (1995), 17–29. <https://doi.org/10.1177/002224379503200104>.
- Itti, Laurent, and Pierre Baldi, "Bayesian Surprise Attracts Human Attention," *Vision Research*, 49 (2009), 1295–1306. <https://doi.org/10.1016/j.visres.2008.09.007>.
- Jehiel, Philippe, "Analogy-Based Expectation Equilibrium," *Journal of Economic Theory*, 123 (2005), 81–104. <https://doi.org/10.1016/j.jet.2003.12.003>.
- Kahneman, Daniel, Jack Knetsch, and Richard Thaler, "Fairness as a Constraint on Profit Seeking: Entitlements in the Market," *American Economic Review*, 76 (1986), 728–741.
- Kahneman, Daniel, and Dan Lovallo, "Timid Choices and Bold Forecasts: A Cognitive Perspective on Risk Taking," *Management Science*, 39 (1993), 17–31. <https://doi.org/10.1287/mnsc.39.1.17>.
- Kahneman, Daniel, and Amos Tversky, "Subjective Probability: A Judgment of Representativeness," *Cognitive Psychology*, 3 (1972), 430–454. [https://doi.org/10.1016/0010-0285\(72\)90016-3](https://doi.org/10.1016/0010-0285(72)90016-3).
- , "Prospect Theory: An Analysis of Decision under Risk," *Econometrica*, 47 (1979), 263–292.
- Kearney, Melissa Schettini, "State Lotteries and Consumer Behavior," *Journal of Public Economics*, 89 (2005), 2269–2299. <https://doi.org/10.1016/j.jpubeco.2004.07.004>.
- Kőszegi, Botond, and Filip Matějka, "Choice Simplification: A Theory of Mental Budgeting and Naive Diversification," *Quarterly Journal of Economics*, 135 (2020), 1153–1207. <https://doi.org/10.1093/qje/qjz043>.
- Kőszegi, Botond, and Adam Szeidl, "A Model of Focusing in Economic Choice," *Quarterly Journal of Economics*, 128 (2013), 53–104. <https://doi.org/10.1093/qje/qjs049>.
- Kunreuther, Howard, *Disaster Insurance Protection: Public Policy Lessons*, (New York: John Wiley & Sons, 1978).
- Laibson, David, "A Cue-Theory of Consumption," *Quarterly Journal of Economics*, 116 (2001), 81–119. <https://doi.org/10.1162/003355301556356>.
- Lanzani, Giacomo, "Correlation Made Simple: Applications to Salience and Regret Theory," *Quarterly Journal of Economics*, 137 (2022), 959–987. <https://doi.org/10.1093/qje/qjab041>.
- Li, Xiaomin, and Colin F. Camerer, "Predictable Effects of Visual Salience in Experimental Decisions and Games," *Quarterly Journal of Economics*, 137 (2022), 1849–1900. <https://doi.org/10.1093/qje/qjac025>.
- Link, Sebastian, Andreas Peichl, Oliver Pfäuti, Christopher Roth, and Johannes Wohlfart, "Attention to the Macroeconomy," CESifo Working Paper No. 10858, 2025. <https://dx.doi.org/10.2139/ssrn.4692410>.
- List, John A., "Neoclassical Theory Versus Prospect Theory: Evidence from the Marketplace," *Econometrica*, 72 (2004), 615–625. <https://www.jstor.org/stable/3598914>.
- , "On the Interpretation of Giving in Dictator Games," *Journal of Political Economy*, 115 (2007), 482–493. <https://doi.org/10.1086/519249>.
- Loewenstein, George, Ted O'Donoghue, and Matthew Rabin, "Projection Bias in Predicting Future Utility," *Quarterly Journal of Economics*, 118 (2003), 1209–1248. <https://doi.org/10.1162/003355303322552784>.

- Ludwig, Jens, and Sendhil Mullainathan, "Machine Learning as a Tool for Hypothesis Generation," *Quarterly Journal of Economics*, 139 (2024), 751–827. <https://doi.org/10.1093/qje/qjad055>.
- Mack, Michael L., and Thomas J. Palmeri, "Discrimination, Recognition, and Classification," in *The Oxford Handbook of Human Memory*, Michael J. Kahana and Anthony D. Wagner, eds. (Oxford: Oxford University Press, 2024), 871–894. <https://doi.org/10.1093/oxfordhb/9780190917982.013.30>.
- Malmendier, Ulrike, and Stefan Nagel, "Learning from Inflation Experiences," *Quarterly Journal of Economics*, 131 (2016), 53–87. <https://doi.org/10.1093/qje/qjv037>.
- Malmendier, Ulrike, Stefan Nagel, and Zhen Yan, "The Making of Hawks and Doves," *Journal of Monetary Economics*, 117 (2021), 19–42. <https://doi.org/10.1016/j.jmoneco.2020.04.002>.
- Manski, Charles, *Identification Problems in the Social Sciences*, (Cambridge, MA: Harvard University Press, 1995).
- McClelland, James L., and David E. Rumelhart, "An Interactive Activation Model of Context Effects in Letter Perception: I. An Account of Basic Findings," *Psychological Review*, 88 (1981), 375–407. <https://doi.org/10.1037/0033-295X.88.5.375>.
- Mohlin, Erik, "Optimal Categorization," *Journal of Economic Theory*, 152 (2014), 356–381. <https://doi.org/10.1016/j.jet.2014.03.007>.
- Mullainathan, Sendhil, "Thinking Through Categories," Working Paper, Harvard University, Cambridge, MA, 2002.
- Mullainathan, Sendhil, Joshua Schwartzstein, and Andrei Shleifer, "Coarse Thinking and Persuasion," *Quarterly Journal of Economics*, 123 (2008), 577–619. <https://doi.org/10.1162/qjec.2008.123.2.577>.
- Nosofsky, Robert M., "Attention, Similarity, and the Identification–Categorization Relationship," *Journal of Experimental Psychology: General*, 115 (1986), 39–57. <https://doi.org/10.1037/0096-3445.115.1.39>.
- Oprea, Ryan, "Decisions under Risk Are Decisions under Complexity," *American Economic Review*, 114 (2024), 3789–3811. <https://doi.org/10.1257/aer.20221227>.
- Rabin, Matthew, and Dimitri Vayanos, "The Gambler's and Hot-Hand Fallacies: Theory and Applications," *Review of Economic Studies*, 77 (2010), 730–778. <https://doi.org/10.1111/j.1467-937X.2009.00582.x>.
- Rabin, Matthew, and Georg Weizsäcker, "Narrow Bracketing and Dominated Choices," *American Economic Review*, 99 (2009), 1508–1543. <https://doi.org/10.1257/aer.99.4.1508>.
- Reed, Stephen K., "Pattern Recognition and Categorization," *Cognitive Psychology*, 3 (1972), 382–407. [https://doi.org/10.1016/0010-0285\(72\)90014-X](https://doi.org/10.1016/0010-0285(72)90014-X).
- Rick, Scott L., Cynthia E. Cryder, and George Loewenstein, "Tightwads and Spendthrifts," *Journal of Consumer Research*, 34 (2008), 767–782. <https://doi.org/10.1086/523285>.
- Rosch, Eleanor, and Barbara Lloyd, *Principles of Categorization*, (London: Routledge, 1978).
- Rubinstein, Ariel, "Similarity and Decision-Making under Risk (Is There a Utility Theory Resolution to the Allais Paradox?)," *Journal of Economic Theory*, 46 (1988), 145–153. [https://doi.org/10.1016/0022-0531\(88\)90154-8](https://doi.org/10.1016/0022-0531(88)90154-8).
- Salant, Yuval, and Ariel Rubinstein, "(A, f): Choice with Frames," *Review of Economic Studies*, 75 (2008), 1287–1296. <https://doi.org/10.1111/j.1467-937X.2008.00510.x>.
- Samuelson, Larry, "Analogies, Adaptation, and Anomalies," *Journal of Economic Theory*, 97 (2001), 320–366. <https://doi.org/10.1006/jeth.2000.2754>.
- Sarkar, Shraman, "Economic Representations," Mimeo, University of Chicago, 2025.
- Savage, Leonard J., "Elicitation of Personal Probabilities and Expectations," *Journal of the American Statistical Association*, 66 (1971), 783–801. <https://doi.org/10.1080/01621459.1971.10482346>.

- Schwartzstein, Joshua, "Selective Attention and Learning," *Journal of the European Economic Association*, 12 (2014), 1423–1452. <https://doi.org/10.1111/jeea.12104>.
- Schwartzstein, Joshua, and Deepak Malhotra, "Rocket Science," Business School Case 921–043, Harvard Business School, Cambridge, MA, 2021.
- Schwartzstein, Joshua, and Adi Sunderam, "Using Models to Persuade," *American Economic Review*, 111 (2021), 276–323. <https://doi.org/10.1257/aer.20191074>.
- Shafir, Eldar, and Richard H. Thaler, "Invest Now, Drink Later, Spend Never: On the Mental Accounting of Delayed Consumption," *Journal of Economic Psychology*, 27 (2006), 694–712. <https://doi.org/10.1016/j.joep.2006.05.008>.
- Shah, Anuj K., Jiaying Zhao, Sendhil Mullainathan, and Eldar Shafir, "Money in the Mental Lives of the Poor," *Social Cognition*, 36 (2018), 4–19. <https://doi.org/10.1521/soco.2018.36.1.4>.
- Simon, Herbert A., "A Behavioral Model of Rational Choice," *Quarterly Journal of Economics*, 69 (1955), 99–118. <https://doi.org/10.2307/1884852>.
- Sims, Christopher A., "Implications of Rational Inattention," *Journal of Monetary Economics*, 50 (2003), 665–690. [https://doi.org/10.1016/S0304-3932\(03\)00029-1](https://doi.org/10.1016/S0304-3932(03)00029-1).
- Stigler, George, and Gary Becker, "De Gustibus Non Est Disputandum," *American Economic Review*, 67 (1977), 76–90.
- Taubinsky, Dmitry, Luigi Butera, Matteo Saccharola, and Chen Lian, "Beliefs About the Economy Are Excessively Sensitive to Household-Level Shocks: Evidence from Linked Survey and Administrative Data," *Quarterly Journal of Economics*, forthcoming.
- Thaler, Richard H., "Mental Accounting Matters," *Journal of Behavioral Decision Making*, 12 (1999), 183–206. [https://doi.org/10.1002/\(SICI\)1099-0771\(199909\)12:3<3C183::AID-BDM318%3E3.0.CO;2-F](https://doi.org/10.1002/(SICI)1099-0771(199909)12:3<3C183::AID-BDM318%3E3.0.CO;2-F).
- , "Toward a Positive Theory of Consumer Choice," *Journal of Economic Behavior & Organization*, 1 (1980), 39–60. [https://doi.org/10.1016/0167-2681\(80\)90051-7](https://doi.org/10.1016/0167-2681(80)90051-7).
- Tversky, Amos, "Elimination by Aspects: A Theory of Choice," *Psychological Review*, 79 (1972), 281–299. <https://doi.org/10.1037/h0032955>.
- , "Features of Similarity," *Psychological Review*, 84 (1977), 327–352.
- Tversky, Amos, and Daniel Kahneman, "Judgment under Uncertainty: Heuristics and Biases," *Science*, 185 (1974), 1124–1131. <https://doi.org/10.1126/science.185.4157.1124>.
- , "The Framing of Decisions and the Psychology of Choice," *Science*, 211 (1981), 453–458. <https://doi.org/10.1126/science.7455683>.
- , "Extensional Versus Intuitive Reasoning: The Conjunction Fallacy in Probability Judgment," *Psychological Review*, 90 (1983), 293–315. <https://doi.org/10.1037/0033-295X.90.4.293>.
- Wachter, Jessica A., and Michael Jacob Kahana, "A Retrieved-Context Theory of Financial Decisions," *Quarterly Journal of Economics*, 139 (2024), 1095–1147. <https://doi.org/10.1093/qje/qjad050>.
- Wakefield, Kirk L., and J. Jeffrey Inman, "Situational Price Sensitivity: The Role of Consumption Occasion, Social Context and Income," *Journal of Retailing*, 79 (2003), 199–212. <https://doi.org/10.1016/j.jretai.2003.09.004>.
- Woodford, Michael, "Prospect Theory as Efficient Perceptual Distortion," *American Economic Review*, 102 (2012), 41–46. <https://doi.org/10.1257/aer.102.3.41>.



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